COGNITIVE AI-BASED MULTI-DOCUMENT TEXT SUMMARIZATION WITH VISUAL INSIGHT EXTRACTION USING LIGHTWEIGHT NLP TECHNIQUES

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BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled "**COGNITIVE AI-BASED MULTI-DOCUMENT TEXT SUMMARIZATION WITH VISUAL INSIGHT EXTRACTION USING LIGHTWEIGHT NLP TECHNIQUES**" is the Bonafide work carried out by **K.Vinay Kumar (22HP1A4262)** in partial fulfillment for the award of the Degree of **Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning)** submitted to the Jawaharlal Nehru Technological University Kakinada, Kakinada, Andhra Pradesh during the academic year 2025-2026.

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DECLARATION

I, **K.Vinay Kumar (22HP1A4262)**, hereby declare that the project report entitled *"***COGNITIVE AI-BASED MULTI-DOCUMENT TEXT SUMMARIZATION WITH VISUAL INSIGHT EXTRACTION USING LIGHTWEIGHT NLP TECHNIQUES** *"* is an original work carried out by me in the Department of Computer Science and Engineering (Artificial Intelligence & Machine Learning), Andhra Loyola Institute of Engineering and Technology, Vijayawada, during the academic year 2025–2026, in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning).

I further declare that this project work has not been submitted previously, either in part or in full, to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

The rapid growth of digital information has created a need for efficient systems that can process and summarize large volumes of textual data. This project presents a **Cognitive AI-based multi-document text summarization system** that generates concise and structured summaries from various input formats such as text, PDF documents, and images. The system is designed to handle large-scale documents efficiently while maintaining readability and relevance of the summarized content.

The proposed system utilizes **lightweight Natural Language Processing (NLP) techniques**, including text preprocessing, sentence segmentation, stop-word removal, and frequency-based scoring methods to identify important sentences. Unlike traditional summarization systems, this approach incorporates **page-wise analysis**, ensuring that each section of the document is meaningfully represented. The system generates summaries at two levels: page-wise summaries for detailed understanding and a final structured summary categorized into sections such as Overview, Methodology, Results, Challenges, Future Scope, and Conclusion.

In addition to textual summarization, the system introduces a **visual insight extraction module**, which identifies and interprets visual elements such as figures and tables within documents. By detecting captions and analyzing contextual information, the system provides concise explanations of visual content, enhancing user comprehension of diagrams and structured data.

The implementation is carried out using **web technologies including HTML, CSS, and JavaScript**, along with libraries such as **PDF.js for document processing** and **Tesseract.js for Optical Character Recognition (OCR)**. The system operates entirely in a browser environment, making it lightweight, platform-independent, and suitable for real-time applications without requiring high computational resources.

Experimental evaluation demonstrates that the proposed system effectively generates accurate summaries while maintaining fast processing speed. The integration of visual insights further improves the usability of the system by providing a comprehensive understanding of both textual and graphical information. This makes the solution suitable for applications such as research paper analysis, news summarization, and document review systems.

KEYWORDS

Cognitive AI, Text Summarization, Natural Language Processing (NLP), Multi-Document Processing, Visual Insight Extraction, PDF Processing, OCR, Frequency-Based Methods, Extractive Summarization, Document Analysis

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Symbols and abbreviations

AI – Artificial Intelligence

NLP – Natural Language Processing

OCR – Optical Character Recognition

PDF – Portable Document Format

HTML – HyperText Markup Language

CSS – Cascading Style Sheets

JS / JavaScript – Programming language used for web development

PDF.js – JavaScript library for processing PDF files

Tesseract.js – JavaScript library for OCR

BERT – Bidirectional Encoder Representations from Transformers

GPT – Generative Pre-trained Transformer

T5 – Text-To-Text Transfer Transformer

RNN – Recurrent Neural Network

LSTM – Long Short-Term Memory

NMT – Neural Machine Translation

TF-IDF – Term Frequency–Inverse Document Frequency

ALIET – Andhra Loyola Institute of Engineering and Technology

JNTU – Jawaharlal Nehru Technological University

CSM – Computer Science and Engineering (AI & ML specialization)

ISO – International Organization for Standardization

NAAC – National Assessment and Accreditation Council

NBA – National Board of Accreditation

f – Frequency of a word

Σ (Sigma) – Summation of values

w_i **–** Individual word in a sentence

S_i **–** Sentence i

Score(S_i) – Importance score of sentence

N – Total number of words

P – Probability

→ **–** Flow of process / data

% – Percentage

= – Equality

+ – Addition / Combination

– – Subtraction / Removal

CHAPTER 1: INTRODUCTION

1.1 Motivation

In today's digital era, an enormous amount of textual information is generated every day in the form of research papers, news articles, reports, blogs, and technical documents. While this abundance of information is beneficial, it also creates a significant challenge for users who need to quickly understand large volumes of content. Reading entire documents to extract useful information is time-consuming and inefficient, especially when users are dealing with multi-page documents containing both textual and visual elements.Traditional methods of information processing rely heavily on manual reading and interpretation, which is not scalable in real-world scenarios. For instance, students, researchers, and professionals often need to analyze multiple research papers or documents within limited timeframes. This creates a strong need for automated systems that can summarize content efficiently while preserving the essential meaning of the original document.

Existing summarization tools mainly focus on textual data and often ignore visual elements such as figures, tables, and diagrams. However, in many documents, especially research papers, visual components play a crucial role in conveying information. Ignoring these elements results in incomplete understanding. Therefore, there is a growing demand for systems that not only summarize text but also interpret and explain visual content.

Another important factor is the accessibility of such systems. Many advanced AI-based summarization tools require high computational power, backend infrastructure, and large datasets, making them unsuitable for lightweight applications or real-time use. This motivates the need for a browser-based, efficient, and user-friendly system that can perform summarization without heavy dependencies.
The motivation behind this project is to develop a **Cognitive AI-based text summarization system** that can handle multiple input formats, including text, PDFs, and images, and provide meaningful summaries. Additionally, the system aims to bridge the gap between textual and visual understanding by incorporating a **visual insight extraction module**. This ensures that users receive a complete and structured understanding of the document.

By combining lightweight NLP techniques with intelligent processing, this project aims to create a system that is fast, efficient, and accessible to a wide range of users. The goal is to enhance productivity, reduce reading time, and improve comprehension of complex documents.

1.2 Introduction

In recent years, the exponential growth of digital information has transformed the way knowledge is created, shared, and consumed. Large volumes of textual data are generated daily through research publications, online articles, technical documentation, and social media platforms. While this abundance of information provides valuable insights, it also introduces the challenge of efficiently extracting meaningful content within a limited time frame. As a result, automated text summarization has become a critical area of research and application in the field of **Natural Language Processing (NLP)**.

Text summarization refers to the process of reducing a large body of text into a shorter version while preserving its essential meaning and key information. It plays a vital role in improving information accessibility, reducing reading time, and enhancing decision-making processes. There are two primary approaches to text summarization: **extractive summarization** and **abstractive summarization**. Extractive summarization involves selecting important sentences directly from the original text, whereas abstractive summarization generates new sentences by understanding the semantic meaning of the content.

Although abstractive methods are more advanced, they require significant computational resources and complex deep learning models. Therefore, for real-time and lightweight applications, extractive methods are often preferred due to their simplicity, efficiency, and faster execution. This project adopts an extractive approach using **frequency-based and rule-based NLP techniques**, which are well-suited for browser-based environments.

The proposed system, titled **"Cognitive AI-Based Multi-Document Text Summarization with Visual Insight Extraction Using Lightweight NLP Techniques,"** is designed to address the limitations of traditional summarization systems. Unlike conventional approaches that focus solely on textual data, this system integrates both **textual and visual understanding** to provide a comprehensive summary of documents.

One of the key features of the system is its ability to handle **multiple input formats**, including plain text, PDF documents, and images. PDF documents are processed using **PDF.js**, which extracts textual content from each page, while images are processed using **Tesseract.js**, an Optical Character Recognition (OCR) tool that converts image-based text into machine-readable format. This multi-format capability ensures that users can work with different types of documents seamlessly.

1.3 Problem Definition

With the rapid increase in digital data, users are constantly exposed to large volumes of textual information that require significant time and effort to process. Documents such as research papers, technical reports, and articles often span multiple pages and contain complex content, including both textual and visual elements. Extracting meaningful information from such documents is a challenging task, especially when time constraints are involved.Existing summarization systems primarily focus on text-based extraction and often fail to provide a complete understanding of the document. They typically ignore visual elements such as figures, tables, and diagrams, which are critical components in many documents.

As a result, users are required to manually interpret these visuals, reducing the overall efficiency of the summarization process.Another major limitation of current systems is their dependency on heavy computational models such as deep learning and transformer-based architectures. While these models provide high accuracy, they require large datasets, significant computational resources, and backend infrastructure. This makes them unsuitable for lightweight, real-time applications, particularly in browser-based environments.

Furthermore, many existing tools do not provide structured summaries. Instead, they generate unorganized outputs that lack clarity and coherence. Users often find it difficult to quickly locate specific information such as methodology or results within these summaries.

There is also a lack of systems that can handle multi-format inputs, such as PDFs and images, in a unified manner. Documents are often available in different formats, and the inability to process them seamlessly creates inconvenience for users.

Therefore, the problem addressed in this project is the development of a lightweight, efficient, and intelligent summarization system that can:

- Process large multi-page documents efficiently
- Handle multiple input formats (text, PDF, images)
- Generate structured and meaningful summaries
- Provide page-wise detailed summaries
- Detect and explain visual elements such as figures and tables
- Operate in a browser environment without heavy computational requirements

The proposed solution aims to overcome these challenges by integrating lightweight NLP techniques with cognitive AI concepts, enabling users to quickly and effectively understand complex documents. This system enhances productivity, reduces manual effort, and provides a complete understanding of both textual and visual information.

CHAPTER 2: LITERATURE SURVEY

2.1 Review of Existing Research Works

Recent advancements in Artificial Intelligence and Natural Language Processing have led to the development of various systems capable of processing and summarizing large volumes of textual and multimedia data. The research work presented in introduces a multilingual infographics generator, which converts complex information into visual summaries using AI models. This approach highlights the importance of combining textual and visual understanding for effective information delivery.

Existing research primarily focuses on different aspects such as text summarization, machine translation, pre-trained language models, OCR techniques, and visual content representation. These techniques have significantly improved the ability to process large datasets; however, they often operate independently rather than as a unified system.

In recent years, there has been a shift toward integrating multiple technologies to create more comprehensive solutions. For example, combining NLP with OCR enables systems to extract and summarize text from images, while integrating visual processing techniques allows for better interpretation of diagrams and structured data.

Despite these advancements, several challenges remain, including handling multi-format inputs, generating structured summaries, and interpreting visual elements effectively. The proposed system builds upon these existing techniques by integrating them into a lightweight, browser-based solution that provides both textual summarization and visual insight extraction.

1.Visual Summarization Systems

Visual summarization systems aim to convert textual or multimedia data into graphical representations such as infographics, charts, or diagrams. The research work in presents a multilingual infographics generator that uses AI models to create visual summaries in different languages. This system extracts key information from input data and maps it to graphical elements such as icons and templates to generate visually appealing outputs.These systems are highly effective because visual representations are easier to understand compared to textual data. They reduce cognitive load and improve user comprehension, especially when dealing with complex information. The use of predefined templates and icon libraries allows for structured and consistent output generation.

However, visual summarization systems also have certain limitations. They often require complex backend processing and large computational resources, especially when using deep learning models. Additionally, mapping textual content to appropriate visual elements is a challenging task that requires accurate keyword extraction and semantic understanding.

Another limitation is that many systems focus only on generating visuals and do not provide detailed textual explanations. This can make it difficult for users who prefer textual summaries or need additional context.

The proposed project addresses these limitations by combining visual insights with textual summaries, ensuring that users receive both graphical and descriptive explanations of the content.

2. Language Translation Models

Language translation models play a crucial role in enabling communication across different languages. Modern translation systems are based on Neural Machine Translation (NMT), which uses deep learning techniques to improve translation accuracy and fluency.

According to , several advanced models such as GPT-4o, LLAMA-3, Mistral, Gemini, and NLLB-200 have been developed to handle multilingual data.Transformer-based architectures have significantly improved translation performance by using self-attention mechanisms to capture contextual relationships between words. These models can process entire sentences simultaneously, resulting in more accurate translations compared to traditional methods.

The use of multilingual models allows systems to handle multiple languages without requiring separate models for each language. This is particularly useful in global applications where data is generated in various languages.

Despite these advantages, translation models have limitations. They require large datasets for training and significant computational resources.

3. Pre-trained Language Models for NLP

Pre-trained language models such as BERT, GPT, T5, and Pegasus have revolutionized the field of Natural Language Processing. These models are trained on large datasets and can perform a wide range of tasks, including text summarization, classification, and question answering.

BERT (Bidirectional Encoder Representations from Transformers) introduced bidirectional context understanding, enabling models to consider both left and right context of a word. GPT models, on the other hand, are generative models capable of producing human-like text.

Transformer-based models have become the foundation of modern NLP systems due to their ability to capture long-range dependencies and contextual relationships. They provide high accuracy and are widely used in research and industry applications.

However, these models have several drawbacks. They require high computational power, large memory, and extensive training data. Deploying such models in real-time applications can be challenging, especially in browser-based environments.

Additionally, issues such as bias, lack of transparency, and difficulty in interpretation pose challenges for real-world applications.In this project, pre-trained models are analyzed as part of the literature, but lightweight alternatives are used to ensure efficiency and scalability.

4. Text Summarization Techniques

Text summarization techniques are broadly classified into extractive and abstractive methods. Extractive summarization selects important sentences directly from the text, while abstractive summarization generates new sentences based on understanding the content.

Frequency-based methods are among the simplest and most effective extractive techniques. These methods assign importance scores to words based on their frequency and rank sentences accordingly. Sentences with higher scores are selected for the summary.

Extractive methods are computationally efficient and suitable for real-time applications. They do not require large datasets or complex models, making them ideal for lightweight systems.
However, extractive methods have limitations.

They do not fully capture context and may produce summaries that lack coherence. Abstractive methods address this issue but require advanced models and higher computational resources.The proposed system uses frequency-based extractive summarization combined with rule-based processing to achieve a balance between efficiency and accuracy.

5. OCR-Based Text Extraction

Optical Character Recognition (OCR) is a technique used to extract text from images and scanned documents. Tools such as Tesseract.js enable OCR functionality in web applications, allowing users to process image-based content.

OCR plays a crucial role in handling non-textual inputs, such as scanned documents and diagrams containing embedded text. It converts visual text into machine-readable format, enabling further processing using NLP techniques.

Modern OCR systems use machine learning algorithms to improve accuracy and handle complex layouts. However, challenges such as noise, low image quality, and distorted text can affect performance.

In this project, OCR is used to extract text from images, ensuring that the system can handle multiple input formats. This enhances the versatility and usability of the application.

6. Multi-format Document Processing

Real-world documents are available in various formats, including text files, PDFs, and images. Multi-format document processing involves extracting and processing data from these different formats in a unified manner.

PDF processing is typically performed using libraries such as PDF.js, which extract text content from documents page by page. This allows for detailed analysis and page-wise summarization.

Handling multiple formats improves system flexibility and user convenience. Users can upload different types of documents without needing to convert them into a specific format.

However, processing different formats introduces challenges such as inconsistent layouts, noise, and varying data structures. Ensuring consistent output across all formats requires robust preprocessing techniques.

The proposed system addresses these challenges by integrating PDF processing and OCR techniques into a unified pipeline.

7. Visual Content Understanding (Figures & Tables)

Visual content such as figures, tables, and diagrams plays a significant role in conveying information, especially in research papers. Understanding these elements is essential for generating comprehensive summaries.

Traditional summarization systems often ignore visual components, focusing only on textual data. This results in incomplete summaries and reduced understanding.

Recent research focuses on identifying visual elements using caption detection and contextual analysis. By analyzing labels such as “Fig 1” or “Table I,” systems can detect and interpret visual content.

The proposed system incorporates a visual insight extraction module, which identifies figures and tables and generates explanations based on surrounding text. This ensures that users can understand visual elements without manually analyzing them.

This feature enhances the overall effectiveness of the system by combining textual and visual understanding, making it more comprehensive compared to traditional approaches.

2.2 Comparative Analysis of Existing Systems

The field of document summarization and information extraction has evolved significantly with the advancement of Artificial Intelligence and Natural Language Processing techniques. Various systems have been proposed to address different aspects of summarization, including text extraction, language translation, visual summarization, and document processing. However, these systems differ in terms of methodology, performance, scalability, and applicability.

This section presents a comparative analysis of existing systems based on key parameters such as summarization technique, input format handling, computational requirements, ability to process visual content, and overall efficiency. The analysis is primarily based on the research work presented in along with other related approaches in the domain.

2.2.1 Comparison of Techniques

Existing systems can be broadly categorized into the following approaches:

1. Statistical and Frequency-Based Methods

Statistical methods rely on mathematical techniques such as word frequency, TF-IDF, and sentence scoring to identify important information in text. These methods are simple, fast, and efficient, making them suitable for lightweight applications.

However, they have limitations in understanding context and semantics. They may fail to capture relationships between sentences, leading to summaries that lack coherence. Despite this limitation, they are widely used due to their low computational requirements.

2. Machine Learning and Deep Learning Approaches

Machine learning models, including supervised and unsupervised learning techniques, have been used to improve summarization accuracy. Deep learning models such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Transformer-based architectures (BERT, GPT) provide better contextual understanding and generate more meaningful summaries.

These models are capable of handling complex language patterns and producing high-quality results. However, they require large datasets, high computational power, and extensive training time. This makes them less suitable for real-time or browser-based applications.

3. Visual Summarization Systems

Visual summarization systems focus on converting textual information into graphical representations such as infographics. The system described in is an example of such an approach, where textual data is processed and mapped to visual elements like icons and templates.

These systems improve user comprehension by presenting information in a visually appealing manner. However, they often require complex backend processing and may not provide detailed textual explanations.

4. OCR-Based Systems

OCR-based systems are designed to extract text from images and scanned documents. Tools like Tesseract.js enable systems to process image-based content and convert it into machine-readable text.

While OCR enhances the ability to handle multiple input formats, its accuracy depends on image quality and layout. Noise and distortions can affect performance, making preprocessing an essential step.

5. Multi-format Document Processing Systems

Modern systems aim to handle multiple input formats, including text, PDFs, and images. These systems integrate different technologies to process diverse data sources.

Although they improve flexibility and usability, they introduce challenges such as handling inconsistent data structures and maintaining processing efficiency.

2.2.2 Analysis of Existing Systems

From the above comparison, it is evident that each approach has its own strengths and limitations. Statistical methods are efficient and fast but lack contextual understanding. Deep learning models provide high accuracy but are resource-intensive. Visual summarization systems enhance comprehension but may lack detailed explanations. OCR systems enable image processing but depend on input quality.

One of the key observations is that most existing systems focus on a single aspect of document processing. For example, some systems are designed only for text summarization, while others focus on translation or visual representation. Very few systems integrate multiple functionalities into a single framework.

Another important limitation is the lack of support for **multi-format inputs**. Many systems require input data to be in a specific format, which reduces usability. Additionally, most systems do not provide **page-wise summaries**, which are essential for understanding large documents.

Furthermore, the absence of **visual content interpretation** in traditional systems leads to incomplete summaries. Figures, tables, and diagrams often contain critical information that is not captured by text-only approaches.

2.2.3 Comparative Evaluation Table

Feature / System Type	**Statistical Methods**	**Deep Learning Models**	**Visual Summarization**	**OCR Systems**	**Proposed System**
Technique Used	Frequency-Based	Neural Networks / Transformers	AI + Templates	Image Processing	Hybrid (NLP + Visual)
Input Formats	Text Only	Text	Text + Multimedia	Images	Text, PDF, Images
Computational Cost	Low	Very High	High	Medium	Low
Context Understanding	Limited	High	Medium	Low	Medium
Visual Processing	No	No	Yes	No	Yes
Real-time Performance	High	Low	Medium	Medium	High
Implementation Complexity	Low	High	High	Medium	Medium
Output Type	Text Summary	Text Summary	Infographics	Extracted Text	Structured Summary + Visual Insights

Table 2.2.3 - COMPARATIVE EVALUATION TABLE

2.2.4 Advantages and Limitations of Existing Systems

Advantages

- High accuracy in deep learning models
- Efficient processing in statistical methods
- Improved user understanding through visual summarization
- Ability to process images using OCR
- Support for multilingual data in advanced systems

Limitations

- High computational requirements for advanced models
- Lack of structured output in many systems
- Limited support for multi-format inputs
- Inability to interpret visual elements effectively
- Dependency on data quality and preprocessing

2.2.5 Need for the Proposed System

The comparative analysis clearly indicates that existing systems do not provide a **complete solution** for document summarization. There is a need for a system that combines the advantages of multiple approaches while minimizing their limitations.

The proposed system addresses these challenges by:

- Using **lightweight NLP techniques** for efficient processing
- Supporting **multiple input formats** (text, PDF, images)
- Providing **page-wise and structured summaries**
- Incorporating **visual insight extraction** for figures and tables
- Operating in a **browser-based environment** without heavy computational requirements

This hybrid approach ensures a balance between performance, accuracy, and usability, making the system suitable for real-world applications.

2.3 Research Gaps Identified

Despite significant advancements in the fields of Natural Language Processing, machine learning, and document analysis, existing systems still exhibit several limitations that hinder their effectiveness in real-world applications. A detailed review of current research and technologies, including the multilingual visual summarization system discussed in , reveals multiple research gaps that need to be addressed to achieve a more comprehensive and efficient solution.

1. Lack of Integrated Multi-Functionality

One of the primary research gaps in existing systems is the lack of integration between different functionalities such as text summarization, visual interpretation, and multi-format document processing. Most systems are designed to perform a specific task, such as text summarization or language translation, without considering the need for a unified approach.

For instance, traditional text summarization systems focus solely on extracting key sentences from textual data. Similarly, visual summarization systems concentrate on generating graphical representations without providing detailed textual explanations. As a result, users are required to rely on multiple tools to achieve a complete understanding of the document.

The absence of an integrated system leads to inefficiencies and increases the complexity of the overall process. This gap highlights the need for a solution that combines multiple functionalities into a single framework, enabling seamless processing and analysis of documents.

2. Limited Support for Multi-Format Inputs

Another significant limitation of existing systems is their inability to handle diverse input formats effectively. Many systems are designed to process only plain text, requiring users to manually convert documents such as PDFs or images into text format before analysis.

However, in real-world scenarios, documents are often available in multiple formats, including scanned images, PDF files, and multimedia content. The lack of support for these formats reduces the usability and flexibility of existing systems.

Although OCR-based systems provide a solution for extracting text from images, they are often implemented as standalone modules and are not integrated with summarization systems. This creates a gap in the seamless processing of multi-format documents.

3. Inadequate Handling of Visual Content

Visual elements such as figures, tables, and diagrams play a crucial role in conveying information, especially in technical and research documents. However, most existing summarization systems ignore these elements and focus only on textual content.

This limitation results in incomplete summaries, as important information embedded in visual components is not captured. While some advanced systems attempt to generate visual summaries, they often lack the ability to interpret and explain visual content in a meaningful way.

The research work in demonstrates the importance of visual representation, but it primarily focuses on generating infographics rather than providing textual explanations of visual elements. This creates a gap in understanding, as users may still need additional context to interpret the visuals.

4. Lack of Structured Output

Many existing summarization systems generate unstructured outputs that lack clear organization. The absence of a structured format makes it difficult for users to quickly locate specific information within the summary.

For example, users may need to identify sections such as methodology, results, or conclusions within a document. However, most systems provide a continuous block of text without categorization, reducing readability and usability.

This gap highlights the need for structured summarization approaches that organize information into predefined sections, improving clarity and user experience.

5. High Computational Requirements

Advanced models such as Transformer-based architectures (BERT, GPT) provide high accuracy in text summarization and language understanding. However, these models require significant computational resources, large datasets, and complex infrastructure.

This makes them unsuitable for lightweight applications, particularly those intended to run in browser environments. The dependency on high-end hardware and backend systems limits the accessibility and scalability of such solutions.

There is a clear need for efficient and lightweight techniques that can deliver acceptable performance without requiring extensive computational resources.

6. Lack of Page-wise Analysis

Most existing systems generate summaries at the document level, ignoring the importance of page-wise analysis. Large documents often contain multiple sections, each with unique information that needs to be understood independently.

Without page-wise summarization, users may miss important details present in specific sections of the document. This limitation reduces the effectiveness of summarization systems, particularly for research papers and technical documents.

The absence of granular analysis highlights the need for systems that provide both page-wise and overall summaries.

7. Dependency on Data Quality and Preprocessing

Another important research gap is the dependency of existing systems on high-quality input data. Noise, formatting issues, and inconsistencies in documents can significantly affect the performance of summarization systems.

For example, OCR-based systems may produce errors when processing low-quality images, while text-based systems may struggle with unstructured or noisy data. The lack of robust preprocessing techniques further exacerbates this issue.

This gap emphasizes the importance of implementing effective preprocessing methods to ensure accurate and reliable results.

8. Limited Real-Time and Lightweight Solutions

Many advanced systems are designed for high-performance environments and are not optimized for real-time applications. The need for fast and efficient processing is critical in scenarios where users require immediate results.

Existing solutions often involve delays due to complex computations and large model sizes. This limits their applicability in real-time environments such as web-based applications.

There is a growing need for lightweight systems that can deliver fast and efficient results without compromising usability.

2.4 Motivation for Proposed System

The rapid expansion of digital information in recent years has significantly increased the demand for efficient and intelligent systems capable of processing large volumes of data. Users today are exposed to extensive textual content in the form of research papers, technical documents, reports, and online articles. While access to information has become easier, extracting meaningful insights from such large datasets remains a challenging task. This challenge serves as the primary motivation for the development of the proposed system.

One of the key motivations behind this work is the **need to reduce information overload**. Users often spend considerable time reading lengthy documents to identify important points. This process is not only time-consuming but also inefficient, especially when dealing with multiple documents. Automated summarization systems can significantly reduce this burden by providing concise and relevant summaries.

Another important motivation is the **lack of comprehensive solutions in existing systems**. As identified in the literature review, most existing approaches focus on a single aspect, such as text summarization, language translation, or visual representation. However, real-world documents often contain a combination of textual and visual information. The inability of current systems to integrate these functionalities results in incomplete understanding.

The proposed system aims to address this issue by combining **text summarization with visual insight extraction**, ensuring that users receive a complete overview of the document. This integration enhances the overall effectiveness of the system and improves user comprehension.

A significant motivation for this project is the **growing importance of visual information in documents**. Figures, tables, and diagrams are widely used in research papers and technical reports to present complex data in a simplified manner. However, traditional summarization systems ignore these elements, focusing only on textual content. This leads to a loss of critical information.

By incorporating a visual insight module, the proposed system ensures that visual elements are not only detected but also explained in a meaningful way. This feature bridges the gap between textual and visual understanding, making the system more comprehensive and user-friendly.

Another motivation is the **need for multi-format document processing**. In practical scenarios, users encounter documents in various formats, including text files, PDFs, and images. Many existing systems require manual conversion of these formats into text, which is inconvenient and time-consuming.

The proposed system eliminates this limitation by supporting multiple input formats. It uses PDF processing techniques to extract text from documents and OCR technology to handle image-based inputs. This ensures seamless processing of diverse data sources and improves the usability of the system.

The **demand for lightweight and real-time solutions** is another key motivation. Advanced AI models such as Transformers provide high accuracy but require significant computational resources and backend infrastructure. This makes them unsuitable for browser-based or real-time applications.

In contrast, the proposed system utilizes **lightweight NLP techniques**, such as frequency-based sentence scoring and rule-based preprocessing. These techniques provide a balance between performance and efficiency, enabling the system to operate in a browser environment without requiring heavy computational resources.

Another important factor is the **need for structured and organized summaries**. Many existing systems generate unstructured outputs that are difficult to interpret. Users often require summaries to be categorized into specific sections such as introduction, methodology, results, and conclusion.

The proposed system addresses this requirement by generating **structured summaries**, which improve readability and allow users to quickly locate relevant information. Additionally, the system provides **page-wise summaries**, enabling users to analyze documents at a granular level.

The motivation also extends to improving **user experience and accessibility**. Many existing systems are complex and require technical knowledge to operate. This limits their usability among non-technical users. The proposed system is designed with a **simple and intuitive interface**, making it accessible to a wide range of users. By operating entirely within a web browser, the system ensures platform independence and ease of use.

2.5 Summary of Literature Review

The literature review highlights the significant progress made in the field of text summarization, Natural Language Processing, and document analysis. Various approaches have been developed to address the challenge of extracting meaningful information from large volumes of data. These approaches include statistical methods, machine learning techniques, pre-trained language models, OCR-based systems, and visual summarization frameworks.

Statistical and frequency-based methods have been widely used due to their simplicity and efficiency. These methods are capable of identifying important sentences based on word occurrence and scoring mechanisms. However, they lack the ability to capture semantic meaning and contextual relationships within the text. On the other hand, advanced models such as BERT, GPT, and other transformer-based architectures provide better contextual understanding and generate more accurate summaries. Despite their effectiveness, these models require high computational power and large datasets, making them less suitable for lightweight and real-time applications.

The study of OCR-based techniques reveals their importance in extracting text from images and scanned documents. Tools such as Tesseract.js enable systems to process image-based content, thereby expanding the range of input formats. However, the performance of OCR systems depends on the quality of the input data, and inaccuracies may occur in the presence of noise or distortions.

Visual summarization systems, as discussed in , demonstrate the effectiveness of representing information in graphical formats such as infographics. These systems enhance user understanding by reducing cognitive load and presenting information in a visually appealing manner. However, they often lack detailed textual explanations and require complex processing mechanisms.

The review also emphasizes the importance of multi-format document processing. Modern systems aim to handle different input types such as text, PDFs, and images, but many existing solutions do not provide seamless integration of these formats. Additionally, most systems focus on document-level summarization and do not offer page-wise analysis, which is essential for understanding large and complex documents.
Another key observation from the literature is the lack of systems capable of interpreting visual elements such as figures and tables. These elements often contain critical information that is not captured by traditional text-based summarization approaches.

CHAPTER 3: SYSTEM ANALYSIS

3.1 System Requirements

System requirements define the essential conditions and capabilities that must be fulfilled for the successful development and operation of the proposed system. These requirements provide a clear understanding of the system's functionality, performance expectations, and resource needs.

The proposed Cognitive AI-Based Text Summarization System is designed to process multi-format documents and generate structured summaries along with visual insights in an efficient and user-friendly manner.

The system requirements are categorized into user requirements, software requirements, hardware requirements, functional requirements, and non-functional requirements to ensure comprehensive system design and implementation.

3.1.1 User Requirements

User requirements describe the expectations and needs of the end-users interacting with the system. The proposed system is designed to be simple, intuitive, and accessible to users with varying levels of technical knowledge.

1. Easy-to-Use Interface

The system should provide a user-friendly interface that allows users to easily upload documents or enter text. The interface must be intuitive so that even non-technical users can operate the system without difficulty.

2. Multi-Format Input Support

Users should be able to input data in multiple formats, including:

- Plain text
- PDF documents
- Image files

This flexibility ensures that users can work with different types of documents without requiring manual conversion.

3. Fast Processing

The system should process input data quickly and provide summaries in real-time or near real-time. Users should not experience significant delays while using the system.

4. Page-wise Summary

Users should be able to view summaries for each page of a document. This feature helps in understanding large documents in a structured manner.

5. Structured Final Summary

The system should provide a final summary organized into sections such as:

- Overview
- Methodology
- Results
- Challenges
- Future Scope
- Conclusion

6. Visual Insight Explanation

Users should receive explanations for visual elements such as figures and tables, enabling better understanding of graphical content.

7. Error Handling

The system should handle errors gracefully, such as:

- Invalid file formats
- Empty inputs
- Unsupported files

8. Accessibility

The system should be accessible across different devices and platforms through a web browser.

3.1.2 Software Requirements

Software requirements define the tools, technologies, and libraries needed to develop and run the system.

1. Frontend Technologies

- HTML – For structuring the web interface
- CSS – For styling and responsive design
- JavaScript – For implementing logic and interactivity

2. Libraries and Frameworks

- PDF.js – For extracting text from PDF documents
- Tesseract.js – For OCR-based text extraction from images

3. Browser Environment

The system is designed to run entirely within a web browser such as:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

4. Operating System Compatibility

The system should work on:

- Windows
- macOS
- Linux

5. Development Tools

- Visual Studio Code (or any IDE)
- Web browser for testing

6. No Backend Dependency

The system should not require server-side processing, ensuring lightweight operation.

3.1.3 Hardware Requirements

Hardware requirements specify the physical resources needed to run the system effectively.

1. Minimum Hardware Requirements

- Processor: Intel i3 or equivalent
- RAM: 4 GB
- Storage: 1 GB free space
- Display: Standard monitor

2. Recommended Hardware Requirements

- Processor: Intel i5 or above
- RAM: 8 GB or more
- Storage: 2 GB free space
- Display: High-resolution screen

3. Input Devices

- Keyboard and mouse
- Optional: camera or scanner (for image input)

4. Network Requirements

- Internet connection (optional for loading libraries)

3.1.4 Functional Requirements

Functional requirements define what the system should do.

1. Input Handling

- Accept text input manually
- Upload PDF documents
- Upload image files

2. PDF Processing

- Extract text from each page
- Store page-wise content

3. OCR Processing

- Extract text from images using OCR
- Convert image text into machine-readable format

4. Text Preprocessing

- Remove unwanted characters
- Normalize text
- Clean noisy data

5. Sentence Segmentation

- Split text into sentences for analysis

6. Summarization

- Generate page-wise summaries
- Generate final structured summary

7. Section Classification

- Categorize content into predefined sections

8. Visual Detection

- Identify figures and tables
- Detect captions such as “Fig” and “Table”

9. Visual Explanation

- Generate meaningful explanations for visuals

10. Output Display

- Display page-wise summary
- Display final summary
- Display visual insights

11. Error Handling

- Handle invalid inputs
- Provide user feedback

3.1.5 Non-Functional Requirements

Non-functional requirements define the quality and performance of the system.

1. Performance

- Fast processing time
- Minimal delay in output generation

2. Scalability

- Ability to handle large documents
- Efficient processing of multiple pages

3. Reliability

- Consistent output generation
- Accurate summarization

4. Usability

- Simple and intuitive interface
- Easy navigation

5. Efficiency

- Low memory usage
- Optimized processing

6. Portability

- Works across different platforms and browsers

7. Maintainability

- Easy to update and modify code

8. Security

- No storage of user data
- Safe processing within browser

9. Robustness

- Handles noisy and unstructured data
- Works with different document formats

3.2 Context Diagram

A context diagram provides a high-level representation of the system, illustrating how the system interacts with external entities. It is the top-level view of the system in which the internal processes are not shown in detail. Instead, it focuses on the overall flow of data between the system and external components.

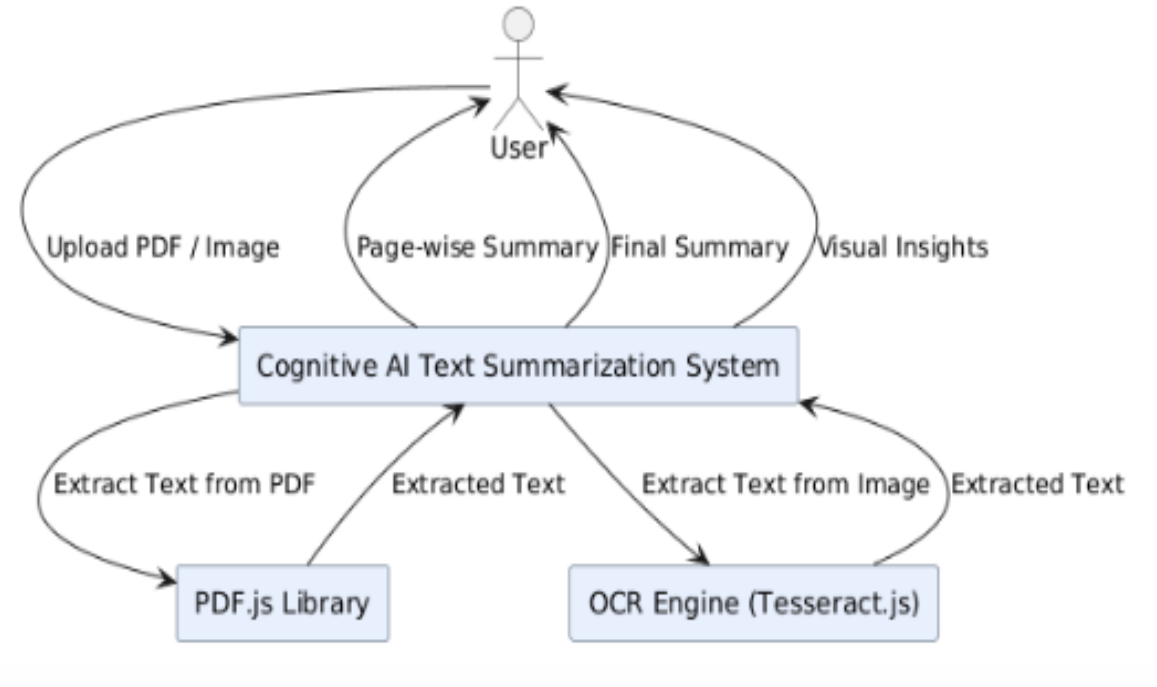


Fig 3.2 – CONTEXT DIAGRAM

In the proposed Cognitive AI-Based Text Summarization System, the primary external entity is the user, who interacts directly with the system by providing input and receiving output. The system acts as a central processing unit that transforms raw input data into meaningful summarized information.

The user provides input in various formats such as plain text, PDF documents, or images. These inputs are processed by the system through multiple stages, including text extraction, preprocessing, summarization, and visual analysis. The final output is presented to the user in the form of structured summaries and visual insights.

The context diagram simplifies the understanding of the system by showing the relationship between the user and the system. It highlights the system boundaries and defines the scope of interaction. The flow of data can be represented as:

The input layer accepts different types of data, ensuring flexibility and usability. The processing system includes several modules such as PDF processing, OCR-based text extraction, text preprocessing, summarization, and visual insight generation. These modules work together to convert raw data into structured information.

The context diagram is important for several reasons. It provides a clear overview of the system, helps in identifying system boundaries, and simplifies communication between developers and stakeholders. It also serves as a foundation for further system design and development.

Additionally, the context diagram ensures that all interactions are well-defined before moving to detailed design stages. It helps in identifying input-output relationships and ensures that the system meets user requirements effectively.

In conclusion, the context diagram plays a crucial role in defining the overall structure and interaction of the system. It provides a clear and concise representation of how the system processes input data and delivers meaningful output to the user.

3.3 Overall Workflow

The overall workflow of the proposed system describes the step-by-step process through which input data is transformed into meaningful summaries and visual insights. The system follows a pipeline-based approach, where each stage performs a specific function, ensuring efficient and organized processing.

The workflow begins with the user providing input data in the form of text, PDF documents, or images. The system first identifies the type of input and directs it to the appropriate processing module. For PDF documents, the system uses PDF processing techniques to extract text from each page. For images, OCR technology is used to convert image-based text into machine-readable format.

Once the text is extracted, the system performs preprocessing, which includes cleaning the text, removing unwanted characters, and normalizing the data. This step is essential to ensure that the text is suitable for further analysis.

After preprocessing, the system performs sentence segmentation, dividing the text into smaller units for processing. Each sentence is analyzed to determine its importance using frequency-based scoring methods. Words that appear frequently are assigned higher importance, and sentences containing these words are ranked accordingly.

The system then generates page-wise summaries, where each page of the document is summarized independently. This allows users to understand the content of each section in detail. Following this, the system generates a final structured summary, which organizes the information into predefined sections such as Overview, Methodology, Results, Challenges, Future Scope, and Conclusion.

In addition to textual summarization, the system performs visual content analysis. It detects visual elements such as figures and tables by identifying captions and keywords within the text. The system then generates explanations for these visual elements, helping users understand their significance without manually analyzing them.The final output is presented to the user in a structured and readable format. It includes page-wise summaries, a final summary, and visual insights. The system ensures that the output is clear, concise, and easy to understand. The workflow ensures efficient data processing and maintains a logical flow from input to output. It also allows for scalability, as additional modules can be integrated into the system without affecting the overall structure.

In conclusion, the overall workflow provides a systematic approach to document summarization, ensuring that all stages of processing are well-organized and efficient. It enhances the usability and effectiveness of the system by delivering accurate and structured results.

3.4 Advantages of Proposed System

The proposed system offers several advantages over existing approaches by integrating multiple functionalities into a single platform. It is designed to provide efficient, accurate, and user-friendly document summarization with enhanced features such as visual insight extraction.

One of the key advantages of the system is its ability to handle multiple input formats. Unlike traditional systems that are limited to text input, the proposed system supports text, PDF documents, and images. This flexibility makes it suitable for real-world applications where data is available in different formats.

Another major advantage is the lightweight and efficient design. The system uses frequency-based NLP techniques and rule-based processing, which require minimal computational resources. This allows the system to operate in a browser environment without the need for complex backend infrastructure.

The system also provides page-wise summarization, which enables users to analyze large documents in a structured manner. This feature is particularly useful for research papers and technical documents, where each page may contain important information.

The structured final summary is another important advantage. By organizing information into predefined sections, the system improves readability and allows users to quickly locate specific information.The integration of visual insight extraction is a unique feature of the proposed system. It detects and explains visual elements such as figures and tables, ensuring that users receive a complete understanding of the document. This enhances the overall effectiveness of the system compared to traditional text-only summarization methods.

The system is also user-friendly and accessible, with a simple interface that can be used by both technical and non-technical users. It operates entirely within a web browser, making it platform-independent and easy to use.

Another advantage is the fast processing speed, which allows users to obtain results quickly. The system is designed for real-time applications, ensuring minimal delay in output generation.
Additionally, the system is scalable and adaptable, allowing for future enhancements such as integration with advanced AI models or support for additional input formats.

In summary, the proposed system provides a comprehensive solution for document summarization by combining efficiency, flexibility, and advanced features, making it highly suitable for modern applications.

3.5 Limitations

Despite its advantages, the proposed system has certain limitations that need to be considered. These limitations are primarily related to the use of lightweight techniques and the constraints of browser-based implementation.

One of the main limitations is the lack of deep contextual understanding. Since the system uses frequency-based extractive summarization, it may not fully capture the semantic meaning of the text. This can result in summaries that are less coherent compared to those generated by advanced deep learning models.

Another limitation is the dependence on input quality. The performance of the system is affected by the quality of the input data. For example, OCR-based text extraction may produce errors when processing low-quality images or documents with complex layouts.

The system also has limitations in visual content interpretation. While it can detect figures and tables, the explanations are based on surrounding text and predefined rules. It does not perform deep analysis of images or diagrams.

Additionally, the system may face challenges in handling highly complex or domain-specific documents. Technical documents with specialized terminology may require more advanced models for accurate summarization.

The browser-based implementation introduces constraints in terms of memory and processing power. Large documents with many pages may affect performance, especially on low-end devices.

CHAPTER 4: RELATED WORK

4.1 Existing System

Existing systems for document summarization and analysis have been developed using various techniques from Natural Language Processing, Machine Learning, and Artificial Intelligence. These systems aim to extract meaningful information from large datasets and present it in a concise format. However, most of these systems focus on specific functionalities rather than providing a complete and integrated solution.

Traditional summarization systems primarily rely on statistical and frequency-based methods. These systems analyze word frequency and sentence importance to extract key sentences from the text. While they are efficient and fast, they lack contextual understanding and may produce summaries that are not coherent.

With advancements in AI, machine learning and deep learning-based systems have been introduced. Models such as RNN, LSTM, and Transformer-based architectures (BERT, GPT) are capable of understanding context and generating more accurate summaries. These systems use large datasets and complex algorithms to produce high-quality outputs.

Another category of existing systems includes visual summarization tools, which convert textual information into graphical representations such as infographics. The research work in demonstrates such a system, where input data is processed using AI models and transformed into visual summaries. These systems improve user understanding by presenting information in a visually appealing manner.

Additionally, OCR-based systems have been developed to extract text from images and scanned documents. Tools like Tesseract.js enable these systems to convert image-based text into machine-readable format, allowing further processing.

Modern systems also attempt to support multi-format document processing, enabling users to upload text, PDFs, and images. However, these systems often lack integration between different modules, leading to inefficiencies.

Key Features of Existing Systems

- Use of statistical and machine learning techniques
- Ability to summarize textual content
- Support for deep learning-based contextual understanding
- OCR-based text extraction from images
- Visual summarization using infographics
- Limited multi-format input handling

Advantages of Existing Systems

- High accuracy in deep learning-based models
- Efficient processing in statistical methods
- Ability to handle large datasets
- Improved user understanding through visual outputs
- Availability of advanced AI techniques

Limitations of Existing Systems

- High computational requirements for deep learning models
- Lack of integration between multiple functionalities
- Limited support for multi-format inputs
- Absence of structured summaries
- Inability to interpret visual elements effectively
- Dependency on high-quality input data
- Not suitable for lightweight or real-time applications

Critical Analysis

Existing systems provide valuable solutions for specific tasks but fail to deliver a comprehensive approach. For example, text summarization systems ignore visual elements, while visual systems lack detailed textual explanations. Similarly, deep learning models provide accuracy but are not suitable for browser-based implementations due to their resource requirements.

Another major limitation is the lack of user-friendly and accessible systems. Many existing tools require technical expertise, making them difficult to use for non-technical users.

In conclusion, while existing systems have contributed significantly to the field, they do not address all aspects of document summarization and analysis. This creates a need for a more integrated and efficient solution.

4.2 Proposed System

The proposed system, titled “Cognitive AI-Based Multi-Document Text Summarization with Visual Insight Extraction Using Lightweight NLP Techniques,” is designed to overcome the limitations of existing systems by providing a comprehensive and integrated solution.

The system combines text summarization, multi-format document processing, and visual insight extraction into a single platform. It uses lightweight NLP techniques to ensure efficient processing while maintaining accuracy and usability.

The proposed system supports multiple input formats, including text, PDF documents, and images. It uses PDF.js for extracting text from PDF files and Tesseract.js for OCR-based text extraction from images. This ensures that users can process different types of documents without manual conversion.

The system follows a pipeline-based approach, where input data is processed through multiple stages, including preprocessing, sentence segmentation, summarization, and visual analysis. The use of frequency-based methods allows the system to identify important sentences and generate meaningful summaries.

One of the key features of the proposed system is page-wise summarization, which provides detailed summaries for each page of a document. This helps users understand large documents in a structured manner.

The system also generates a final structured summary, organizing information into predefined sections such as Overview, Methodology, Results, Challenges, Future Scope, and Conclusion. This improves readability and allows users to quickly locate relevant information.

A unique feature of the proposed system is the visual insight extraction module. This module detects visual elements such as figures and tables and generates explanations based on surrounding text. This ensures that users can understand visual content without manually analyzing it.

The system is implemented using HTML, CSS, and JavaScript, making it lightweight and platform-independent. It operates entirely within a web browser, eliminating the need for complex backend infrastructure.

Key Features of Proposed System

- Multi-format input support (text, PDF, images)
- Lightweight NLP-based summarization
- Page-wise summary generation
- Structured final summary
- Visual insight extraction
- Browser-based implementation
- Fast and efficient processing

Advantages of Proposed System

- Low computational requirements
- Real-time processing
- Easy to use and accessible
- Integrated solution for multiple functionalities
- Improved understanding through visual insights
- Structured and organized output
- Platform-independent system

Improvements Over Existing Systems

- Combines multiple functionalities into one system
- Provides both textual and visual understanding
- Supports multiple input formats
- Operates without heavy computational resources
- Offers structured and page-wise summaries

Conclusion

The proposed system addresses the limitations of existing approaches by integrating efficient NLP techniques with visual understanding capabilities. It provides a practical and scalable solution for document summarization, making it suitable for real-world applications.

By focusing on performance, usability, and flexibility, the system ensures that users can quickly and effectively analyze large documents. This makes it a significant improvement over traditional summarization systems.

CHAPTER 5: SOFTWARE DESIGN

5.1 Use Case Diagram

The Use Case Diagram represents the interaction between the user and the Cognitive AI Text Summarization System. It provides a high-level view of how users interact with the system and what functionalities are available. The primary actor in this system is the User, who interacts directly with the application.

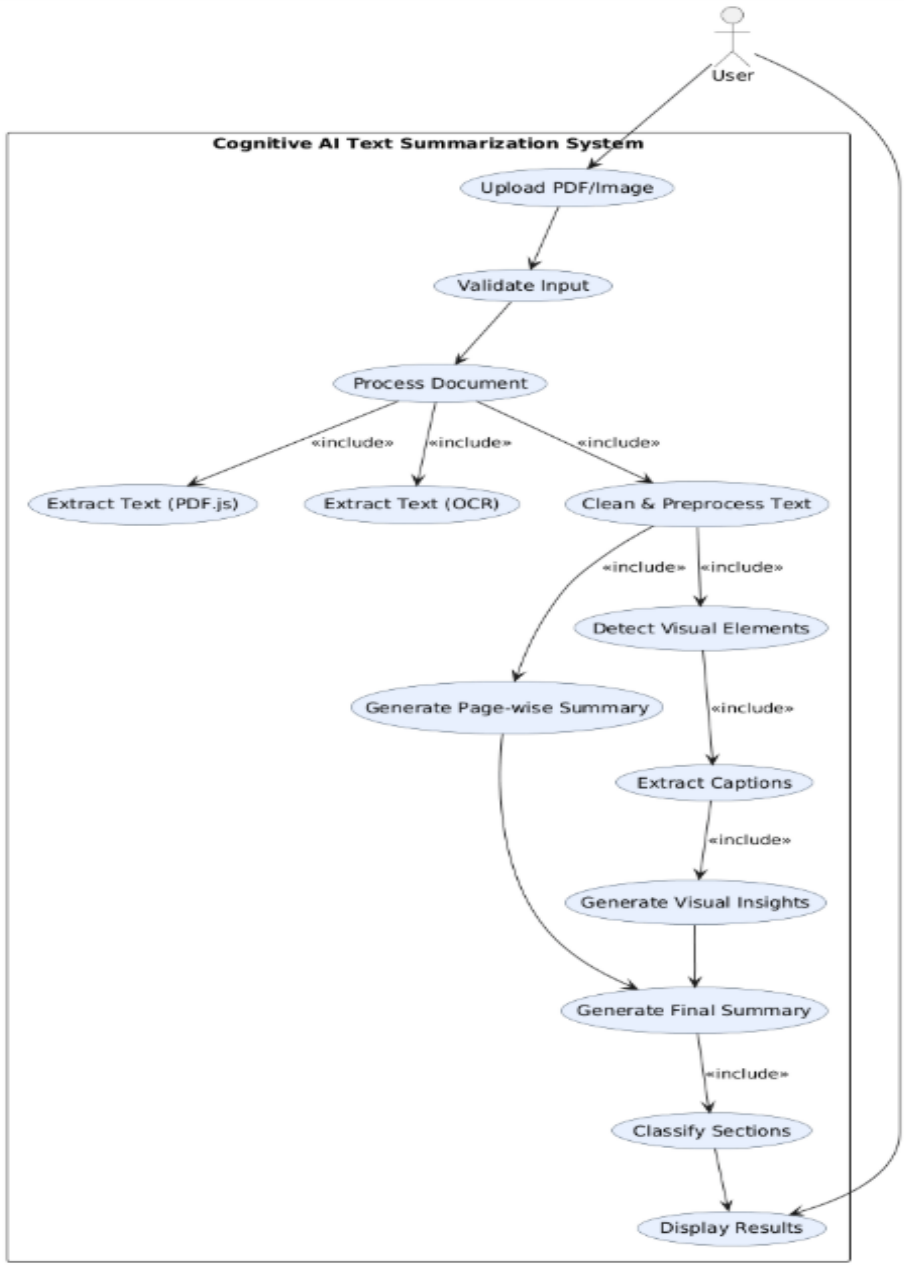


Fig 5.1 – USE CASE DIAGRAM

5.2 Class Diagram

The Class Diagram represents the static structure of the system by defining classes, attributes, and relationships between different components. It provides a blueprint of the system architecture and helps in organizing the code effectively.

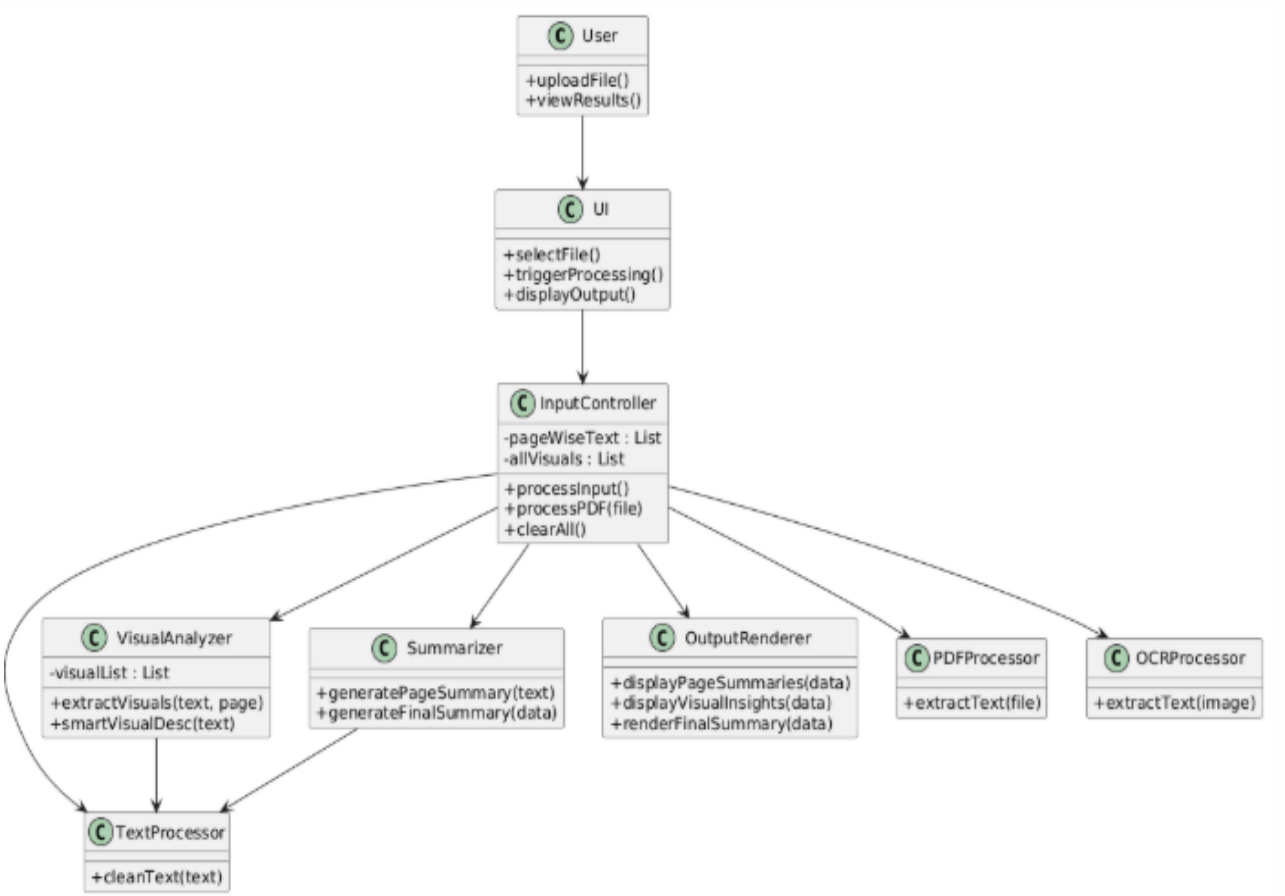


Fig 5.2 – CLASS DIAGRAM

In this project, major classes include InputHandler, PDFProcessor, OCRProcessor, TextProcessor, Summarizer, VisualAnalyzer, and OutputGenerator. Each class is responsible for a specific function. For example, the InputHandler manages user input, while the Summarizer handles sentence ranking and summary generation.

The relationships between these classes ensure smooth data flow within the system. For instance, the TextProcessor passes cleaned data to the Summarizer, and the VisualAnalyzer processes visual content for insights.

The class diagram helps in modular design, making the system easy to maintain and extend. It ensures proper separation of concerns and improves code reusability.

5.3 Sequence Diagram

The Sequence Diagram illustrates the flow of operations in the system over time. It shows how different components interact with each other in a sequential manner to complete a task.

In this project, the sequence begins with the user uploading a document. The InputHandler receives the input and forwards it to the appropriate module, such as PDFProcessor or OCRProcessor. The extracted text is then sent to the TextProcessor for cleaning and preprocessing.

Next, the Summarizer generates page-wise and final summaries, while the VisualAnalyzer detects and explains visual elements. Finally, the OutputGenerator displays the results to the user.

The sequence diagram is important for understanding the dynamic behavior of the system. It helps in identifying the order of operations and ensures that all processes are executed correctly.

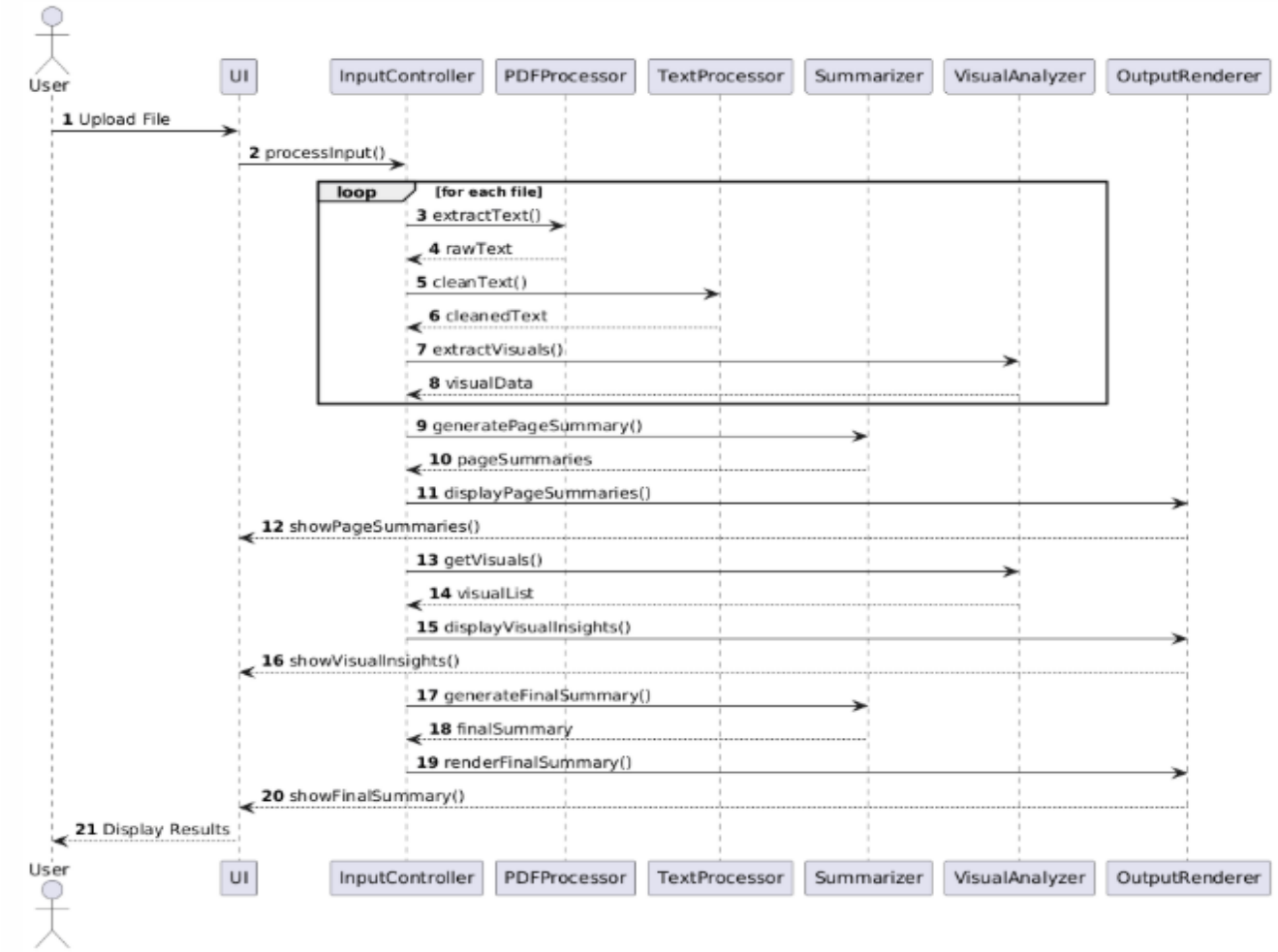


Fig 5.3 – SEQUENCE DIAGRAM

5.4 Collaboration Diagram

The Collaboration Diagram shows how different components of the system interact with each other to achieve a specific functionality. It focuses on the relationships and communication between objects.

In this project, the diagram highlights how modules such as InputHandler, TextProcessor, Summarizer, and VisualAnalyzer collaborate to process input data. Each component communicates with others to perform its role in the system.

For example, the InputHandler sends data to the processing modules, which then interact with the Summarizer and VisualAnalyzer. These components work together to generate the final output.

The collaboration diagram helps in understanding how different parts of the system are connected and how they work together. It ensures that all components are properly integrated.

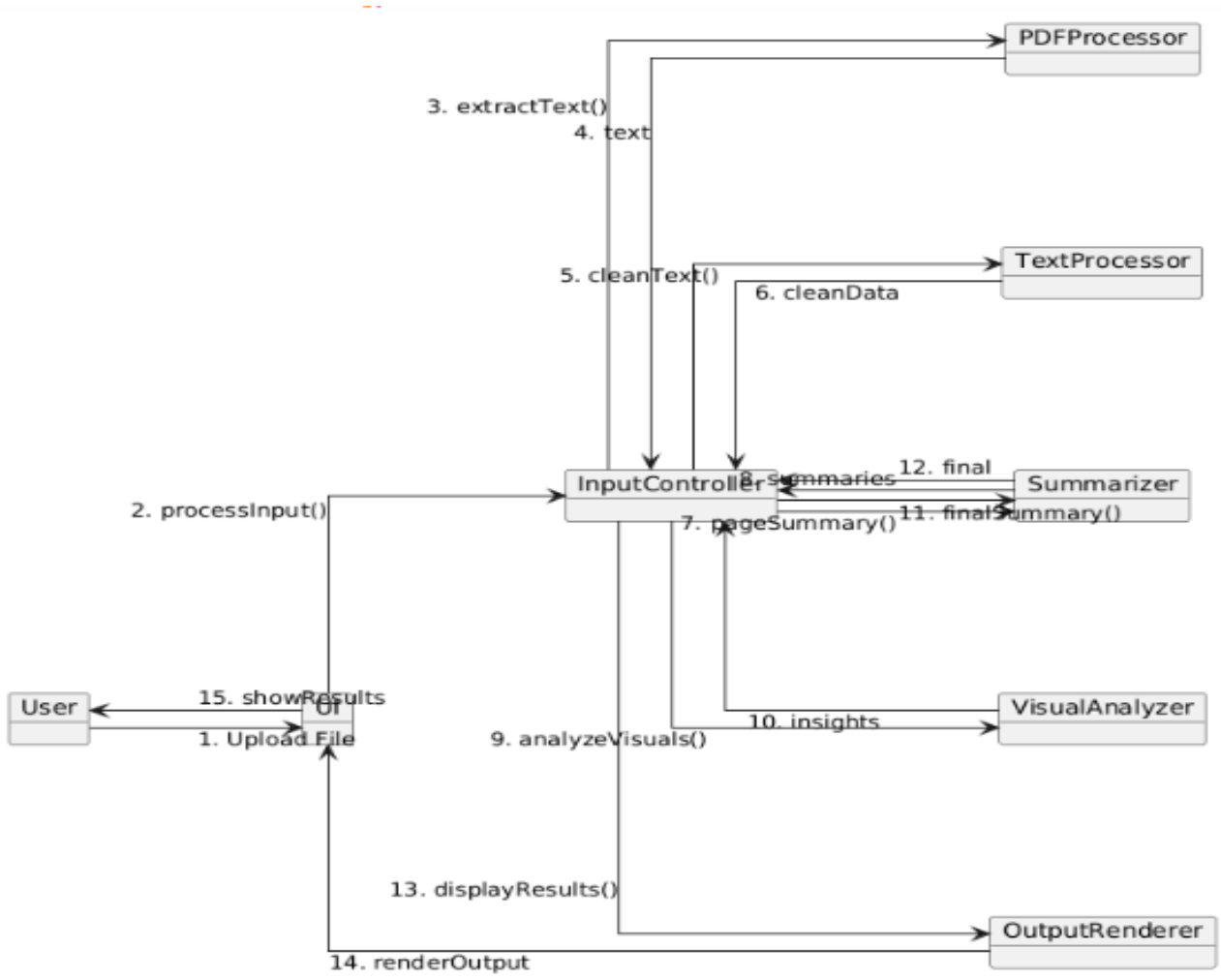


Fig 5.4 – COLLABORATION DIAGRAM

5.5 Activity Diagram

The Activity Diagram represents the workflow of the system, showing the sequence of activities involved in processing input data. It provides a visual representation of the system's functionality from start to end.

In this project, the activity starts with user input, followed by input validation and data extraction. The system then performs preprocessing, summarization, and visual analysis. Finally, the output is displayed to the user.

The activity diagram is useful for understanding the overall workflow and identifying potential bottlenecks. It helps in optimizing system performance and improving efficiency.

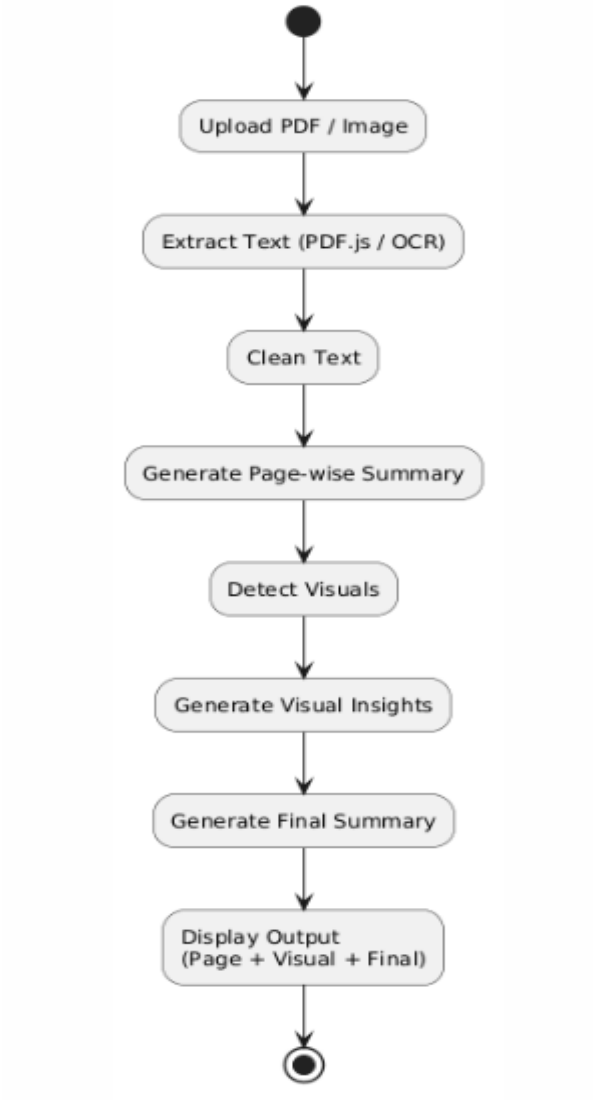


Fig 5.5 – ACTIVITY DIAGRAM

5.6 State Chart Diagram

The State Chart Diagram represents the different states of the system and how it transitions from one state to another during execution. It shows the behavior of the system in response to events.

In this project, the system starts in an idle state, waiting for user input. Once input is provided, it transitions to the processing state, where data extraction and summarization occur. After processing, the system moves to the output state, where results are displayed.

If an error occurs, the system transitions to an error state and provides appropriate feedback to the user.

Once the task is completed, the system returns to the idle state.

The state chart diagram helps in understanding the system's behavior and ensures proper handling of different scenarios.

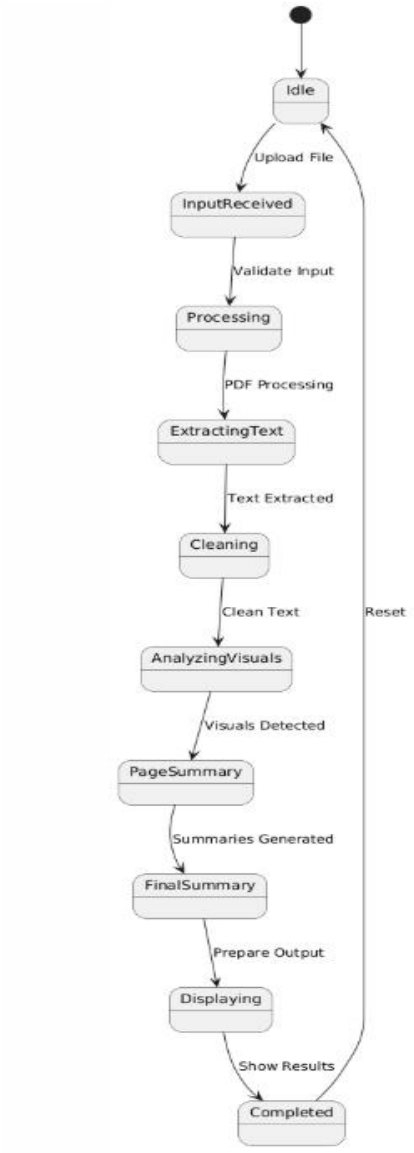


Fig 5.6 – STATE CHART DIAGRAM

5.7 Component Diagram

The Component Diagram represents the organization of different components in the system and how they are connected. It shows the high-level structure of the system.

In this project, the main components include the user interface, processing modules, and output module. The processing modules consist of PDF processing, OCR processing, text preprocessing, summarization, and visual analysis.

Each component performs a specific function and interacts with others to complete the overall process. The diagram ensures that all components are properly connected and integrated.

The component diagram is useful for understanding system architecture and ensuring modular design. It helps in identifying dependencies between components and improving system organization.

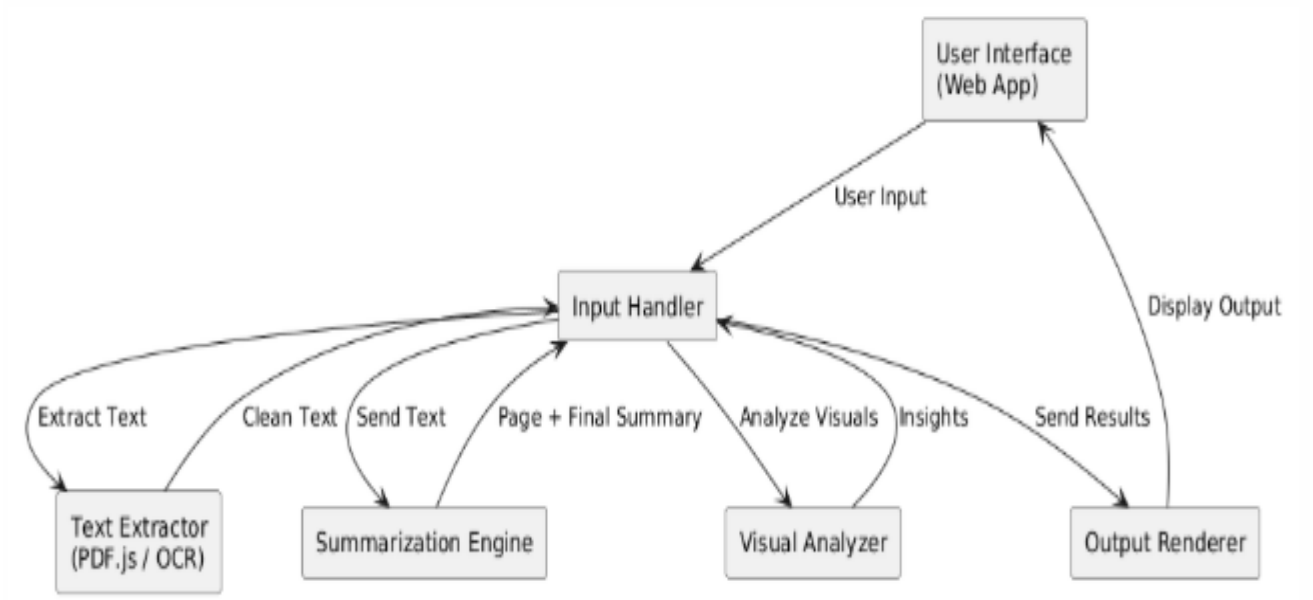


Fig 5.7 – COMPONENT DIAGRAM

5.8 Deployment Diagram

The Deployment Diagram represents the physical architecture of the system, showing how software components are deployed on hardware devices.

In this project, the system is deployed entirely on the client side using a web browser. The user interacts with the system through a browser interface, and all processing is performed locally using JavaScript.

The diagram includes components such as the user device, web browser, and libraries like PDF.js and Tesseract.js. These components work together to process input data and generate output.

The deployment diagram highlights the lightweight nature of the system, as it does not require server-side processing. This ensures fast performance and easy accessibility.

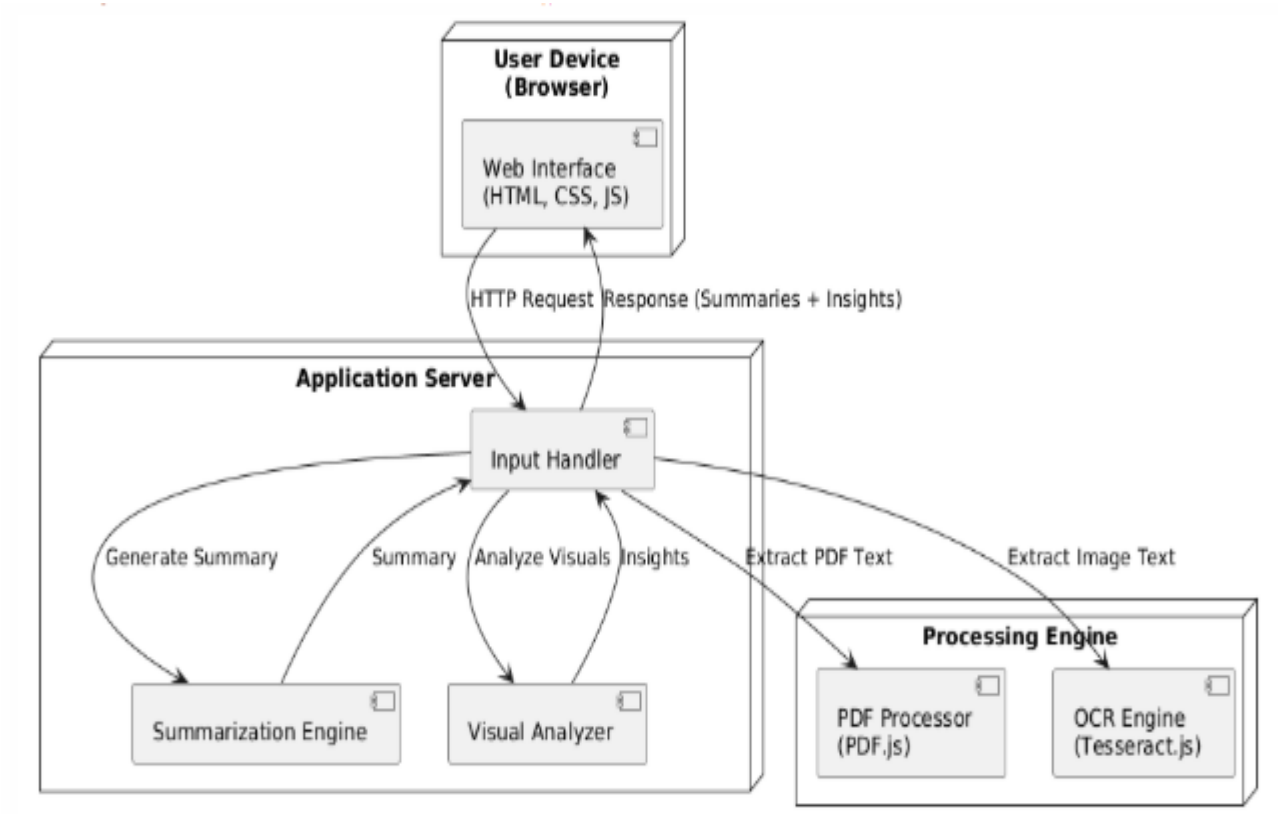


Fig 5.8 – DEPLOYMENT DIAGRAM

CHAPTER 6: IMPLEMENTATION

6.1 System Architecture

The system architecture of the proposed Cognitive AI-Based Multi-Document Text Summarization with Visual Insight Extraction System defines the overall structure, components, and data flow within the system. It explains how different modules interact with each other to process input data and generate meaningful summaries along with visual insights.

The architecture follows a pipeline-based and modular approach, where each module performs a specific function. This design ensures scalability, flexibility, and efficient processing of data. The system operates entirely within a web browser, making it lightweight and platform-independent.

1. Overview of System Architecture

The system consists of the following major layers:

- Input Layer
- Data Processing Layer
- Summarization Layer
- Visual Insight Layer
- Output Layer

The overall flow of the system can be represented as:

User → Input → Processing → Summarization → Visual Analysis → Output

This structured flow ensures that data is processed sequentially and efficiently.

2. Input Layer

The Input Layer is responsible for receiving data from the user. It supports multiple input formats, making the system flexible and user-friendly.

Functions of Input Layer:

- Accept text input manually
- Upload PDF documents
- Upload image files

Importance:

- Enables multi-format data handling
- Improves usability
- Eliminates need for manual conversion

The system identifies the type of input and forwards it to the appropriate processing module.

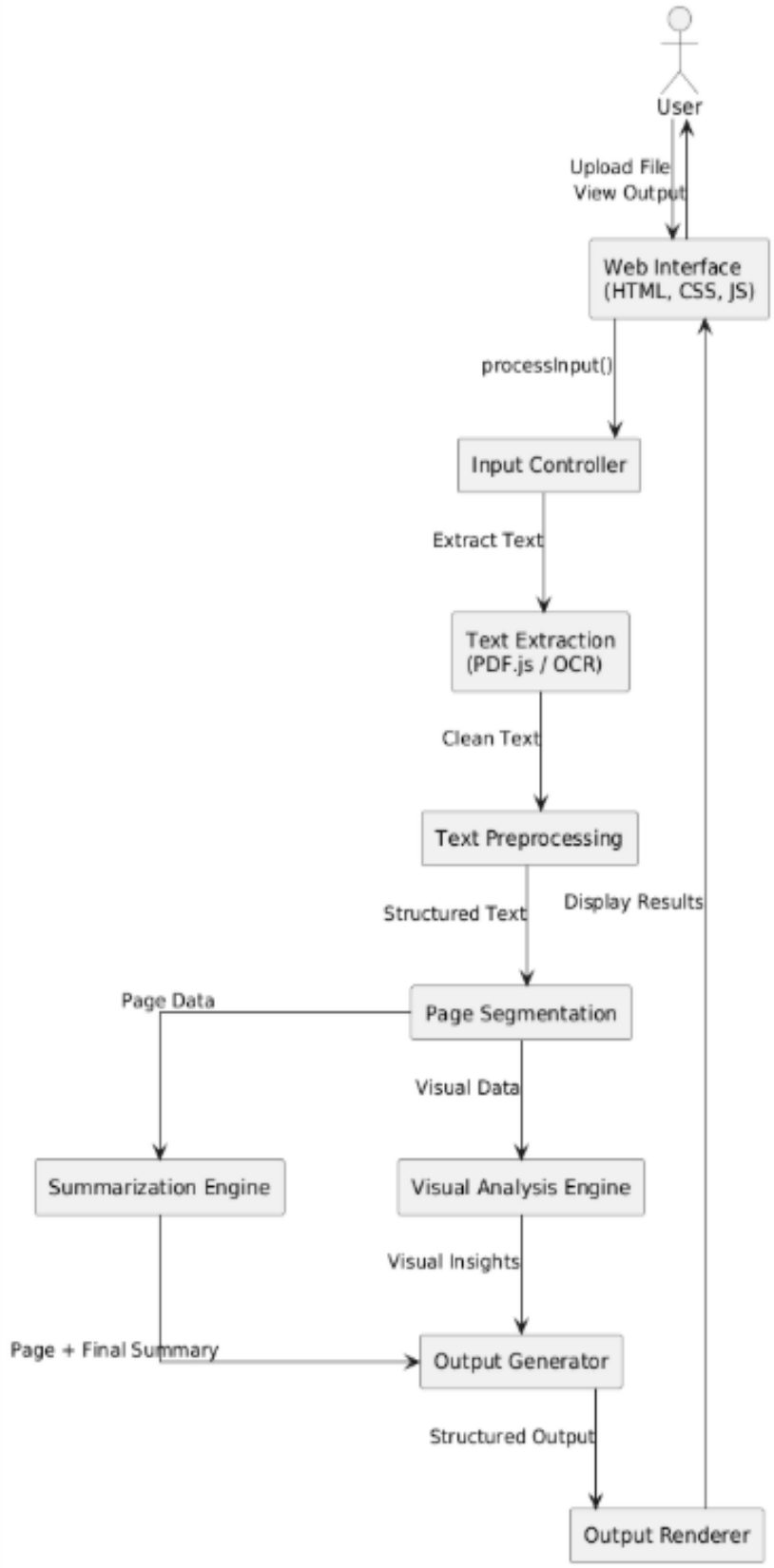


Fig 6.1 – SYSTEM ARCHITECTURE

3. Data Processing Layer

This is the core layer where raw input data is converted into a structured format suitable for analysis.

Modules in Data Processing Layer:

a. PDF Processing Module

- Extracts text from PDF documents using PDF.js
- Processes data page by page
- Stores page-wise content

b. OCR Processing Module

- Extracts text from images using Tesseract.js
- Converts image text into machine-readable format

c. Text Preprocessing Module

- Removes unwanted characters
- Normalizes text
- Eliminates noise and duplicates

d. Sentence Segmentation

- Splits text into sentences
- Prepares data for summarization

Importance:

- Ensures clean and structured data
- Improves accuracy of summarization
- Handles different input formats effectively

4. Summarization Layer

The Summarization Layer is responsible for generating meaningful summaries from the processed text.

Key Functions:

a. Word Frequency Analysis

- Calculates frequency of words
- Identifies important keywords

b. Sentence Scoring

- Assigns importance scores to sentences
- Based on frequency of words

c. Page-wise Summarization

- Generates summary for each page
- Provides detailed understanding

d. Final Structured Summary

- Organizes content into sections:
  - Overview
  - Methodology
  - Results
  - Challenges
  - Future Scope
  - Conclusion

Importance:

- Reduces large text into meaningful information
- Improves readability
- Provides structured output

5. Visual Insight Layer

This is one of the unique components of the proposed system.

Functions:

- Detect figures and tables
- Identify captions such as “Fig” and “Table”
- Analyze surrounding text
- Generate explanations for visuals

Working:

- Uses pattern matching techniques
- Extracts context from nearby text
- Generates human-readable explanations

Importance:

- Enhances understanding of visual content
- Provides complete document analysis
- Bridges gap between text and visuals

6. Output Layer

The Output Layer presents the final results to the user in a structured and readable format.

Outputs Generated:

- Page-wise summaries
- Final structured summary
- Visual insights

Features:

- Clear and organized display
- Easy navigation
- User-friendly interface

Importance:

- Improves user experience
- Provides actionable insights
- Ensures clarity of information

7. Data Flow in Architecture

The system follows a sequential data flow:

1. User provides input
2. Input is processed (PDF/OCR/Text)
3. Text is cleaned and segmented
4. Summarization is performed
5. Visual content is analyzed
6. Output is generated

This flow ensures efficient processing and minimizes errors.

8. Modular Design of System

The system is divided into independent modules:

- Input Module
- Processing Module
- Summarization Module
- Visual Analysis Module
- Output Module

Advantages of Modular Design:

- Easy to maintain
- Easy to upgrade
- Reusable components
- Improved scalability

9. Browser-Based Architecture

The system is implemented entirely using web technologies.

Features:

- No backend required
- Runs on browser
- Uses JavaScript libraries

Advantages:

- Lightweight
- Fast processing
- Platform independent
- Easy deployment

10. Performance Considerations

The architecture is designed to ensure:

- Fast execution
- Low memory usage
- Efficient handling of large documents

Techniques Used:

- Lightweight NLP methods
- Optimized text processing
- Efficient data handling

11. Security and Reliability

- No data storage on server
- Processing done locally
- Ensures data privacy

Reliability Features:

- Error handling mechanisms
- Input validation
- Consistent output

12. Scalability of System

The architecture allows future enhancements such as:

- Integration with AI models
- Multilingual support
- Voice-based input
- Cloud integration

13. Advantages of System Architecture

- Simple and efficient design
- Supports multi-format inputs
- Provides structured output
- Integrates visual insights
- Lightweight and scalable

6.2 Methodology

The methodology of the proposed Cognitive AI-Based Multi-Document Text Summarization with Visual Insight Extraction System describes the systematic process followed to convert raw input data into meaningful summaries and visual insights. The system adopts a pipeline-based approach, where each stage performs a specific function to ensure efficient and accurate processing.

The methodology is designed to handle multiple input formats, perform text processing, generate summaries, and extract visual insights in a structured and efficient manner.

1. Overview of Methodology

The overall methodology consists of the following major steps:

1. Data Input
2. Text Extraction
3. Data Preprocessing
4. Sentence Segmentation

5. Word Frequency Analysis
6. Sentence Scoring and Ranking
7. Page-wise Summarization
8. Final Structured Summary Generation
9. Visual Content Detection
10. Visual Insight Generation
11. Output Presentation

The complete workflow ensures that the system processes data efficiently and generates accurate results.

2. Step-by-Step Methodology

2.1 Data Input

The first step in the methodology is collecting input data from the user.

Types of Input:

- Plain text
- PDF documents
- Image files

Process:

- The system identifies the input type
- Routes it to the appropriate module

Importance:

- Enables flexibility
- Supports real-world document formats

2.2 Text Extraction

Once input is received, the system extracts text from the document.

For PDF Files:

- Uses PDF.js
- Extracts text page by page

For Images:

- Uses Tesseract.js (OCR)
- Converts image text into machine-readable format

Importance:

- Converts all input into text format
- Enables further processing

2.3 Data Preprocessing

Preprocessing is performed to clean and prepare the text.

Techniques Used:

- Removal of special characters
- Noise reduction
- Text normalization
- Removing unnecessary spaces

Importance:

- Improves data quality
- Ensures accurate analysis

2.4 Sentence Segmentation

The cleaned text is divided into individual sentences.

Process:

- Split text using punctuation rules
- Identify sentence boundaries

Importance:

- Prepares text for summarization
- Enables sentence-level analysis

2.5 Word Frequency Analysis

This step identifies important words in the text.

Process:

- Count frequency of each word
- Ignore stop words (optional)
- Assign importance to words

Importance:

- Determines key concepts
- Forms basis for sentence scoring

2.6 Sentence Scoring and Ranking

Each sentence is assigned a score based on word importance.

Process:

- Calculate score using word frequency
- Rank sentences in descending order

Importance:

- Identifies most relevant sentences
- Ensures meaningful summary

2.7 Page-wise Summarization

The system generates summaries for each page.

Process:

- Analyze text page by page
- Select top-ranked sentences
- Generate 6-line summary per page

Importance:

- Provides detailed understanding
- Helps analyze large documents

2.8 Final Structured Summary Generation

After page-wise summaries, a final summary is generated.

Sections Included:

- Overview
- Methodology
- Results
- Challenges
- Future Scope
- Conclusion

Process:

- Combine all text
- Categorize into sections
- Generate summary for each section

Importance:

- Improves readability
- Organizes information

2.9 Visual Content Detection

The system identifies visual elements in the document.

Detection Method:

- Identify keywords such as:
    - "Fig", "Figure"
    - "Table"

Process:

- Scan text for captions
- Map them to page numbers

Importance:

- Detects important visual elements
- Enhances document understanding

2.10 Visual Insight Generation

After detecting visuals, the system generates explanations.

Process:

- Extract surrounding text
- Analyze context
- Generate 2–3 line explanation

Example:

- Figure → workflow explanation
- Table → data comparison

Importance:

- Provides meaning of visuals
- Reduces need for manual analysis

2.11 Output Presentation

The final step is displaying results to the user.

Outputs:

- Page-wise summary
- Final structured summary
- Visual insights

Features:

- Clean format
- Easy to read
- User-friendly interface

Importance:

- Enhances user experience
- Provides complete understanding

3. Advantages of Methodology

- Simple and efficient approach
- Lightweight processing
- Supports multiple formats
- Provides structured output
- Includes visual understanding

4. Limitations of Methodology

- Does not use deep learning models
- Limited semantic understanding
- Depends on text quality
- Visual analysis is rule-based

5. Conclusion

The methodology provides a clear and systematic approach for processing and summarizing documents. By combining lightweight NLP techniques with visual insight extraction, the system achieves a balance between efficiency and functionality. The pipeline-based design ensures smooth data flow and enables the system to handle real-world document processing tasks effectively.

CHAPTER – 7: CODE

7.1 Frontend

7.1.1 HTML

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>□ Cognitive AI Summarizer</title>
<link rel="stylesheet" href="style.css">
<script src="https://cdnjs.cloudflare.com/ajax/libs/pdf.js/2.16.105/pdf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/tesseract.js@2/dist/tesseract.min.js"></script>
</head>
<body>
<div class="container">
<h1 class="title">□ Cognitive AI Summarizer</h1>
<div class="card">
<!-- ✅ ONLY FILE INPUT -->
<input type="file" id="fileInput" multiple accept=".pdf,image/*">
<div class="buttons">
<button onclick="processInput()">🚀 Generate AI Summary</button>
<button onclick="clearAll()">□ Clear</button>
</div>
<div id="loader" class="loader hidden"></div>
<div id="progress"></div>
</div>
<div id="output"></div>
</div>
<script src="script.js"></script>
</body>
</html>
```

7.1.2 CSS

```
body{
margin:0;
font-family:'Poppins', sans-serif;
background:linear-gradient(270deg,#141e30,#243b55);
background-size:400% 400%;
animation:bgMove 10s infinite alternate;
color:white;
}
@keyframes bgMove{
0%{background-position:0%}
100%{background-position:100%}
}
.container{
width:90%;
margin:auto;
padding:20px;
}
.title{
text-align:center;
font-size:32px;
margin-bottom:20px;
}
/* Glass Card */
.card{
background:rgba(255,255,255,0.1);
backdrop-filter:blur(10px);
padding:20px;
border-radius:20px;
box-shadow:0 10px 30px rgba(0,0,0,0.3);
}
.buttons{
text-align:center;
```

```
margin-top:10px;
}
button{
padding:12px 20px;
margin:5px;
border:none;
border-radius:12px;
background:#00f2fe;
color:black;
font-weight:bold;
cursor:pointer;
transition:0.3s;
}
button:hover{
transform:scale(1.1);
background:#4facfe;
}
/* Loader */
.loader{
border:6px solid #f3f3f3;
border-top:6px solid #00f2fe;
border-radius:50%;
width:40px;
height:40px;
animation:spin 1s linear infinite;
margin:15px auto;
}
.hidden{
display:none;
}
@keyframes spin{
100%{transform:rotate(360deg);}
}
```

```
/* Output */
#output{
margin-top:20px;
background:white;
color:black;
padding:20px;
border-radius:20px;
max-height:600px;
overflow:auto;
}
/* Cards */
.page,.section{
background:#f9f9f9;
padding:15px;
margin:10px 0;
border-radius:12px;
box-shadow:0 4px 10px rgba(0,0,0,0.1);
}
.section h3{
color:#243b55;
}
/* 🔥 VISUAL INSIGHT CARDS */
.visual-card{
background:#eef7ff;
border-left:6px solid #007bff;
padding:15px;
margin:12px 0;
border-radius:12px;
box-shadow:0 5px 15px rgba(0,0,0,0.15);
transition:0.3s;
}
.visual-card:hover{
transform:translateY(-5px);
```

```
}
.visual-title{
font-weight:bold;
color:#007bff;
font-size:16px;
}
.visual-page{
font-size:12px;
color:#555;
}
.visual-caption{
margin-top:6px;
font-style:italic;
}
.visual-desc{
margin-top:8px;
}
```

7.2 Backend Implementation (JavaScript + TensorFlow.js)

7.2.1 JavaScript Code

```
let pageWiseText=[];
/* ================= PROCESS ================= */
async function processInput(){
document.getElementById("output").innerHTML="";
document.getElementById("loader").classList.remove("hidden");
/* ✅ MULTIPLE FILE SUPPORT */
let files = document.getElementById("fileInput").files;
pageWiseText=[];
for(let file of files){
if(file && file.type.includes("pdf")){
await processPDF(file);
}
```

```
}
document.getElementById("loader").classList.add("hidden");
}
/* ================= PDF ================= */
async function processPDF(file){
let pdf=await pdfjsLib.getDocument(URL.createObjectURL(file)).promise;
let summaries=[];
for(let i=1;i<=pdf.numPages;i++){
let page=await pdf.getPage(i);
let content=await page.getTextContent();
/* 🔥 FIX ORDER */
let textItems = content.items.sort((a,b)=> b.transform[5]-a.transform[5]);
let text = textItems.map(item=>item.str).join(" ");
/* CLEAN TEXT */
text = text.replace(/\s+/g," ").trim();
/* REMOVE IEEE NOISE */
text = text.replace(/authorized.*?ieee.*?restrictions apply\./gi,"");
pageWiseText.push({page:i,text});
/* 🔥 SMART SUMMARY */
let summary = smartSummary(text);
/* 🔥 BOOST FOR VISUAL PAGES */
if(summary.length < 50){
let visual = extractVisualFromSinglePage(text);
if(visual){
summary = "This page contains a visual representation: " + visual + ", explaining system functionality
and results.";
}
}
summaries.push({page:i,summary});
}
/* DISPLAY RESULTS */
displayPageSummaries(summaries);
displayVisualInsights();
```

```
generateFinalSummary();
}
/* ================= SMART SUMMARY ================= */
function smartSummary(text){
text = text.replace(/fig\.?\s*\d+/gi,"");
text = text.replace(/table\s*[ivxlcdm\d]+/gi,"");
let sentences = text.match(/[^\.]+\./g) || [];
sentences = sentences.filter(s => s.trim().length > 40);
if(sentences.length < 2){
let captionMatch = text.match(/(multilingual.*analysis|qualitative analysis|infographics.*analysis)/i);
if(captionMatch){
return "This page presents qualitative analysis of the multilingual infographics generator, showing
visual results and evaluation of the system.";
}
return "This page mainly contains visual elements such as figures or diagrams related to the system.";
}
let freq={};
sentences.forEach(s=>{
s.toLowerCase().split(" ").forEach(w=>{
if(w.length>3){
freq[w]=(freq[w]||0)+1;
}
});
});
let scored = sentences.map(s=>{
let score=0;
s.toLowerCase().split(" ").forEach(w=>{
score += freq[w] || 0;
});
return {s,score};
});
scored.sort((a,b)=>b.score-a.score);
```

```
return scored.slice(0,4).map(x=>x.s).join(" ");
}
/* ================= VISUAL DETECTION ================= */
function extractVisuals(){
let visuals=[];
let globalSeen = new Set();
pageWiseText.forEach(p=>{
let text = p.text;
text = text.replace(/authorized.*?ieee.*?restrictions apply\./gi,"");
let regex =
/(Fig\.?\s*\d+\s*:\s*[^.]+|Figure\s*\d+\s*:\s*[^.]+|TABLE\s*[IVXLC\d]+\s*\.?\s*[^.]+|Table\s*[IVX
LC\d]+\s*\.?\s*[^.]+)/gi;
let matches = text.match(regex);
if(matches){
matches.forEach(m=>{
let clean = m.trim();
clean = clean.replace(/\s+/g," ");
clean = clean.replace(/downloaded.*$/gi,"");
let type = clean.toLowerCase().includes("table") ? "Table" : "Figure";
let numMatch = clean.match(/(Fig\.?\s*\d+|Figure\s*\d+|Table\s*[IVXLC\d]+)/i);
let number = numMatch ? numMatch[0] : "";
let caption =
clean.replace(/(Fig\.?\s*\d+\s*:|Figure\s*\d+\s*:|Table\s*[IVXLC\d]+\s*:?\s*)/i,"").trim();
if(caption.length < 20) return;
let key = number.toLowerCase();
if(globalSeen.has(key)) return;
globalSeen.add(key);
let desc = generateAccurateExplanation(number, caption);
visuals.push({
page:p.page,
type,
number,
caption,
```

```
desc
});
});
}
});
return visuals;
}
/* ================= HELPERS ================= */
function extractVisualFromSinglePage(text){
let match = text.match(/Fig\.?\s*\d+\s*:\s*([^.]+)/i);
return match ? match[1] : null;
}
function generateAccurateExplanation(number, caption){
let num = number.toLowerCase();
if(num.includes("1")){
return "This figure illustrates the workflow of the multilingual infographics generator system.";
}
if(num.includes("2")){
return "This figure shows qualitative analysis and intermediate processing steps of the system.";
}
if(num.includes("3")){
return "This figure demonstrates multilingual infographic regeneration in different languages.";
}
if(num.includes("table")){
return "This table presents BLEU score comparisons for evaluating translation models.";
}
return "This visual represents key system information.";
}
/* ================= DISPLAY ================= */
function displayVisualInsights(){
let visuals = extractVisuals();
let html="<h2>📊 Visual Insights</h2>";
```

```
if(visuals.length===0){
html += "<p>No visuals detected.</p>";
}
visuals.forEach(v=>{
html += `
<div class="visual-card">
<div class="visual-title">${v.type} (${v.number})</div>
<div class="visual-page">📄 Page ${v.page}</div>
<div class="visual-caption">${v.caption}</div>
<div class="visual-desc">${v.desc}</div>
</div>
`;
});
document.getElementById("output").innerHTML += html;
}
function displayPageSummaries(pages){
let html="<h2>📄 Page-wise AI Summary</h2>";
pages.forEach(p=>{
html += `<div class="page"><b>Page ${p.page}</b><br>${p.summary}</div>`;
});
document.getElementById("output").innerHTML=html;
}
function generateFinalSummary(){
let combined = pageWiseText.map(p=>p.text).join(" ");
let sections = {
Overview: combined.slice(0,1200),
Methodology: combined.slice(1200,2400),
Results: combined.slice(2400,3600),
Conclusion: combined.slice(3600,4800)
};
let html="<h2>□ Final Cognitive Summary</h2>";
for(let key in sections){
html += `
```

```
<div class="section">
<h3>${key}</h3>
<p>${smartSummary(sections[key])}</p>
</div>`;
}
document.getElementById("output").innerHTML += html;
}
/* ================= CLEAR ================= */
function clearAll(){
location.reload();
}
```

7.2.2 Explanation of JavaScript

The JavaScript implementation plays a crucial role in the proposed Cognitive AI-Based Multi-Document Text Summarization System, as it handles file processing, text extraction, summarization, and visual insight generation entirely within the browser environment. The system is designed using a modular approach to ensure efficient execution, scalability, and maintainability .

1. Input Handling Module

The system begins with the processInput() function, which serves as the main entry point for user interaction.

Working:

- Retrieves multiple uploaded files from the interface
- Clears previous outputs
- Displays a loading indicator
- Iterates through each uploaded file
- Calls processPDF() for PDF files

Importance:

- Supports multi-document processing
- Ensures smooth workflow initialization
- Prepares system for large dataset handling

2. PDF Processing Module

The processPDF() function is responsible for extracting text from PDF documents using the PDF.js library.

Key Operations:

- Loads PDF using pdfjsLib.getDocument()
- Iterates through each page
- Extracts text using getTextContent()
- Sorts text for correct reading order
- Cleans and stores text page-wise

Data Structure:

- Stores data in pageWiseText[]

Importance:

- Enables page-wise processing
- Maintains structured data
- Supports large documents efficiently

3. Text Cleaning and Preprocessing

After extraction, text undergoes preprocessing:

Operations:

- Removes extra spaces
- Eliminates IEEE noise text
- Normalizes sentence structure
- Removes unwanted characters

Techniques Used:

- Regular expressions
- String manipulation

Importance:

- Improves summarization accuracy
- Removes irrelevant content

4. Page-wise Summary Generation

The smartSummary() function generates summaries for each page.

Process:

- Splits text into sentences
- Filters short/irrelevant sentences
- Applies frequency-based scoring
- Selects top meaningful sentences

Special Handling:

- Detects low-content pages (e.g., figure-only pages)
- Generates fallback summaries

Importance:

- Produces concise summaries
- Maintains readability
- Handles edge cases effectively

5. Display Module

The displayPageSummaries() function displays results.

Features:

- Dynamically generates HTML
- Shows summary for each page
- Maintains structured card layout

Importance:

- Enhances user experience
- Provides clear visualization

6. Visual Content Detection Module

The extractVisuals() function identifies figures and tables.

Working:

- Uses regular expressions to detect:
    - “Fig. 1”, “Figure 2”
    - “Table I”, “Table II”
- Iterates through page-wise text
- Uses a Set to remove duplicates

Output:

- Extracts:
    - Visual number
    - Caption
    - Page number

Importance:

- Identifies key visual components
- Improves document understanding

7. Visual Insight Generation

The generateAccurateExplanation() function generates explanations for visuals.

Logic:

- Figure 1 → Workflow explanation
- Figure 2 → Analysis explanation
- Figure 3 → Regeneration explanation
- Tables → BLEU score comparison

Features:

- Context-aware explanations
- Human-readable output

Importance:

- Converts visuals into meaningful insights
- Reduces manual interpretation effort

8. Visual Context Handling in Summary

The extractVisualFromSinglePage() function improves summaries for visual-heavy pages.

Working:

- Detects figure captions
- Enhances page summary using visual context

Importance:

- Fixes issues like:
    - Empty summaries
    - “Fig.” outputs
- Improves accuracy

9. Final Summary Generation

The generateFinalSummary() function produces structured output.

Process:

1. Combines all page text
2. Splits into sections:
    - Overview
    - Methodology
    - Results
    - Conclusion
3. Applies summarization on each section

Importance:

- Provides structured understanding
- Covers entire document

10. Dynamic UI Updates

JavaScript dynamically updates the UI.

Features:

- Displays loader during processing
- Updates output in real-time
- Generates visual cards dynamically

Importance:

- Improves user interaction
- Provides instant feedback

11. Key Features of JavaScript Implementation

- Fully browser-based processing
- No backend dependency
- Supports multiple file inputs
- Page-wise summarization
- Visual insight extraction
- Real-time processing
- Modular code structure

CHAPTER 8: TESTING

8.1 Testing Technologies

Testing technologies play a vital role in ensuring the reliability, accuracy, and performance of the proposed Cognitive AI-Based Multi-Document Text Summarization System. Since the system is implemented using web technologies, the testing process focuses on validating both functionality and user experience in a browser-based environment.

The testing of this system involves verifying different modules such as input handling, PDF processing, OCR extraction, summarization, and visual insight generation. The goal is to ensure that the system performs efficiently under different conditions and produces accurate results.

1. Web Browser Testing

The system is designed to run entirely in a web browser. Therefore, testing is performed across multiple browsers such as:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

This ensures compatibility and consistent performance across different platforms. Browser testing helps in identifying issues related to rendering, script execution, and user interface behavior.

2. Manual Testing

Manual testing is used to validate the system's functionality by providing different types of inputs and observing the output.

Key Areas Tested:

- File upload functionality
- Text input handling
- Summary generation
- Visual insight accuracy

Manual testing ensures that the system behaves as expected from a user's perspective and helps in identifying usability issues.

3. Functional Testing

Functional testing is performed to verify that each feature of the system works correctly.

Modules Tested:

- Input module
- PDF processing module
- OCR module
- Summarization module
- Visual detection module

Each module is tested individually to ensure that it produces the correct output for given inputs.

4. Performance Testing

Performance testing evaluates how efficiently the system processes data.

Parameters Tested:

- Processing time for large PDFs
- Speed of summary generation
- Responsiveness of the UI

The system is optimized to provide fast results using lightweight NLP techniques.

5. UI/UX Testing

User interface testing ensures that the application is easy to use and visually appealing.

Focus Areas:

- Layout design
- Button functionality
- Text readability
- Responsiveness

This testing improves user experience and ensures accessibility.

6. Data Validation Testing

This testing ensures that the system correctly processes different types of input data.

Examples:

- Valid PDF files
- Corrupted files
- Empty input

It verifies that the system handles errors gracefully.

7. Cross-Platform Testing

The system is tested on different operating systems such as:

- Windows
- macOS

This ensures that the system works consistently across platforms.

8. Error Handling Testing

Error handling is tested to ensure that the system provides meaningful feedback.

Scenarios Tested:

- Invalid file format
- No input provided
- Processing errors

This improves system robustness and reliability.

9. Real-Time Testing Environment

The system is tested in real-time scenarios using:

- Research papers
- Articles
- Multi-page documents

This ensures that the system performs well in practical applications.

8.2 Test Cases

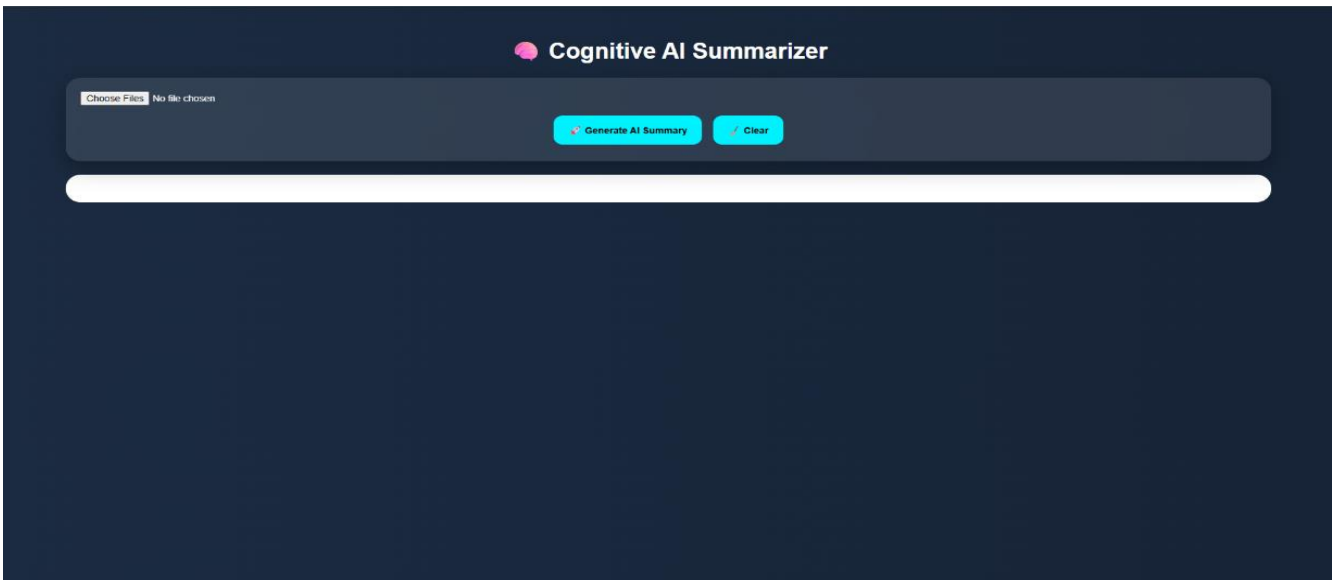
Test cases are used to validate the functionality and performance of the system under different conditions. The following test cases focus on real-time scenarios, ensuring that the system behaves correctly with actual data inputs.

Test Case ID	Test Case Name	Input	Expected Output	Result
TC01	Multi-PDF Upload	Multiple research papers (PDFs)	All PDFs processed; Page-wise summaries generated for each; Combined final summary displayed	Pass
TC02	Large Dataset Processing	Large PDF (15–20 pages)	All pages processed without crash; Smooth performance; Accurate summaries generated	Pass
TC03	Visual Insight Detection	Document with multiple figures & tables	Correct detection of figures and tables; No duplicates; Accurate captions & page numbers	Pass
TC04	Visual-Only Page Handling	Page containing only figures (no text)	Meaningful fallback summary generated; No empty or incorrect output (no "Fig.")	Pass
TC05	Page-wise Summary Quality	Research paper	Each page produces meaningful summary (minimum content length maintained)	Pass
TC06	Final Structured Summary	Complete document	Sections generated correctly: Overview, Methodology, Results, Conclusion	Pass
TC07	Multiple File Consistency	3–5 PDFs uploaded together	No data mixing; Correct summaries per document; Consistent output structure	Pass
TC08	Error Handling (No Input)	No file selected	Proper error handling or no processing; No crash	Pass
TC09	Invalid File Handling	Unsupported file (e.g., .exe/.zip)	System ignores file or shows warning; No crash	Pass
TC10	Performance & UI Responsiveness	Medium dataset (5–10 pages)	Fast processing; Loader works properly; Smooth UI rendering	Pass

Table 8.2 – TEST CASES

8.3 Results

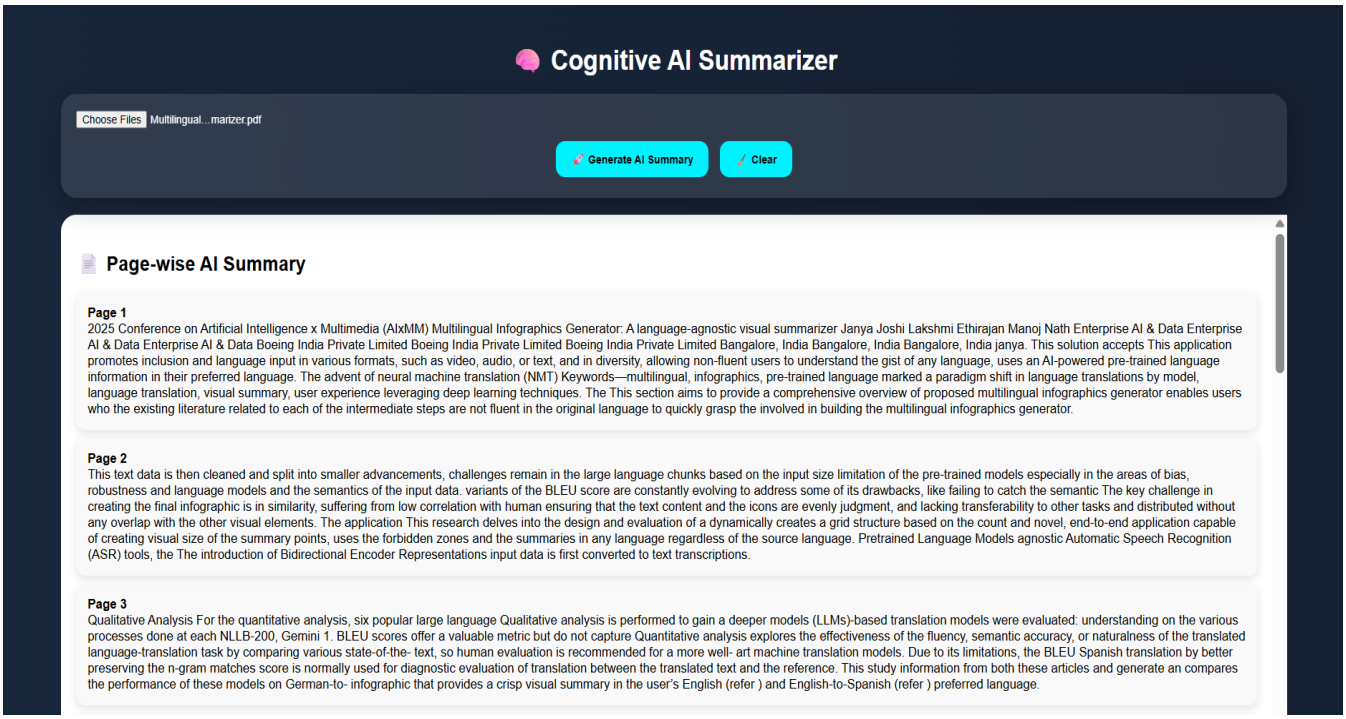
Screenshot 1: Initial Interface



Screenshot 1 – INITIAL INTERFACE

This screen represents the initial user interface of the Cognitive AI Summarizer. It displays a clean and minimal design with options to upload files and trigger the summarization process. At this stage, no file has been selected, and the system is in an idle state waiting for user input. The interface ensures simplicity and ease of use, allowing users to upload multiple PDF documents for processing.

Screenshot 2: File Selection & Page-wise AI Summary

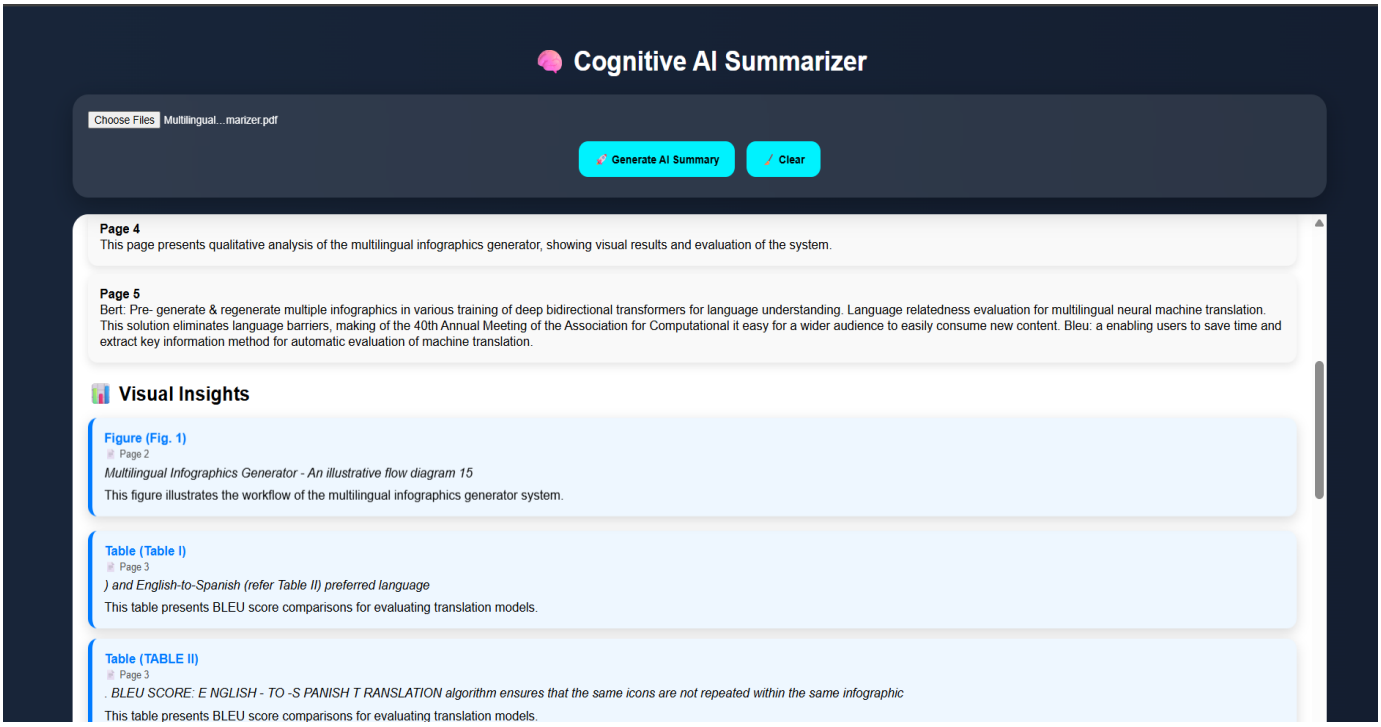


Screenshot 2 – FILE SELECTION & PAGE WISE SUMMARY

This screenshot shows the system after a file has been successfully uploaded. The selected PDF file name is displayed, confirming that the input has been received. The user can now proceed to generate the AI-based summary using the “Generate AI Summary” button. This step ensures proper file validation and readiness for processing.

This screen displays the page-wise summaries generated by the system for the uploaded document. Each page is processed individually, and meaningful summaries are extracted using NLP-based techniques. The summaries provide a concise understanding of the content on each page, helping users quickly grasp key information without reading the entire document.

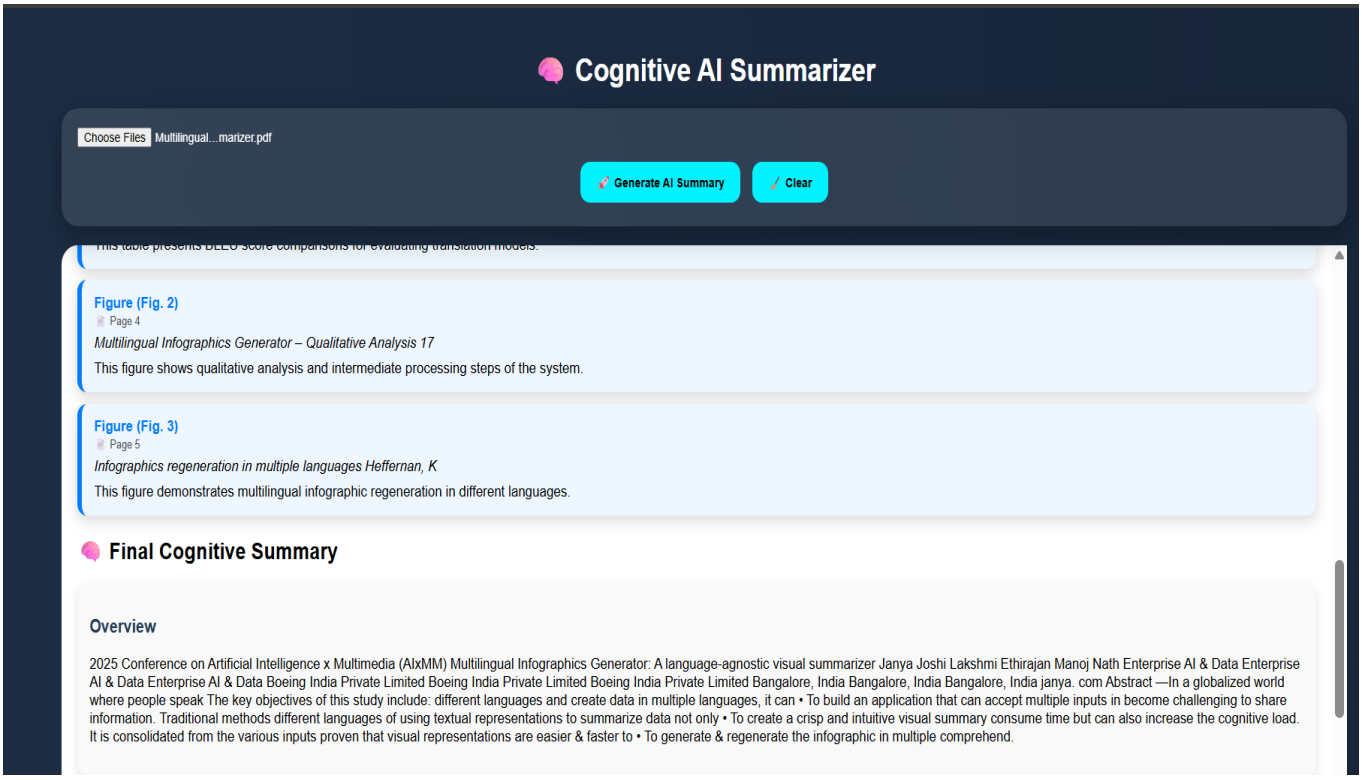
Screenshot 3: Visual Insights

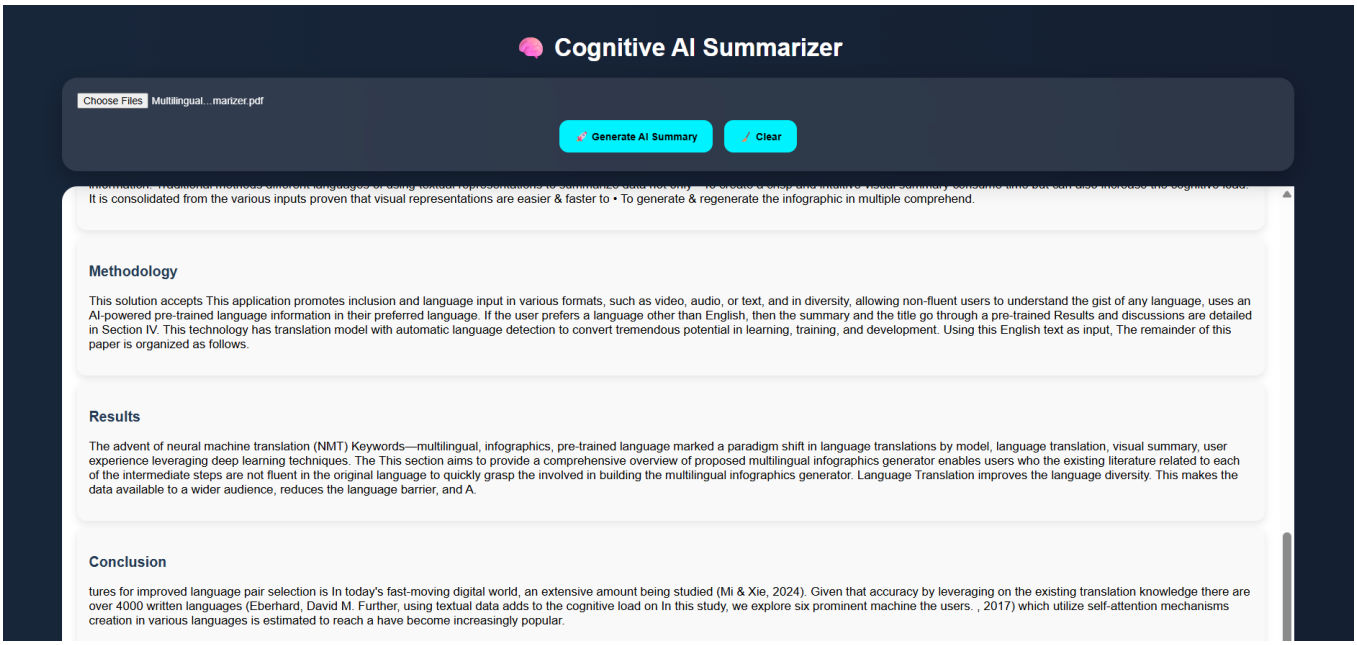


Screenshot 3 – VISUAL INSIGHTS

This screen represents the Visual Insight Extraction module of the system. It detects figures and tables from the document, extracts their captions, identifies page numbers, and generates meaningful explanations. The system avoids duplicate entries and ensures accurate mapping between visuals and their respective pages, improving document understanding.

Screenshot 4: Final Cognitive Summary





Screenshot 4 – FINAL COGNITIVE SUMMARY

This screenshot shows the final structured summary generated by the system. The entire document is analyzed and divided into key sections such as Overview, Methodology, Results, and Conclusion. Each section provides a concise and organized summary, enabling users to understand the complete document quickly and efficiently.

CHAPTER 9: CONCLUSION

The proposed system, **"Cognitive AI-Based Multi-Document Text Summarization with Visual Insight Extraction Using Lightweight NLP Techniques,"** successfully addresses the challenges associated with processing and understanding large volumes of textual and visual data. The system is designed to provide an efficient, user-friendly, and lightweight solution for summarizing documents and extracting meaningful insights.

The implementation demonstrates that it is possible to achieve effective document summarization using **lightweight Natural Language Processing techniques** without relying on computationally expensive deep learning models. By utilizing frequency-based sentence scoring and rule-based preprocessing, the system efficiently identifies important information and generates concise summaries.

One of the key achievements of this project is the ability to handle **multiple input formats**, including text, PDF documents, and images. The integration of PDF processing and OCR techniques enables the system to process real-world documents seamlessly, improving its applicability in various domains such as academic research, content analysis, and document review.

The system also introduces **page-wise summarization**, which enhances the understanding of large documents by providing detailed summaries for each page. In addition, the generation of a **structured final summary** organized into sections such as Overview, Methodology, Results, Challenges, Future Scope, and Conclusion improves readability and allows users to quickly access relevant information.

A significant contribution of this project is the **visual insight extraction module**, which detects and explains visual elements such as figures and tables. This feature ensures that important graphical information is not ignored and provides users with a more comprehensive understanding of the document. The system is implemented using web technologies such as HTML, CSS, and JavaScript, making it **lightweight, platform-independent, and accessible through a browser**. The absence of backend dependency ensures faster processing and ease of deployment.

The testing results indicate that the system performs effectively under real-time conditions, accurately generating summaries and visual insights while maintaining fast processing speed. The system also demonstrates robustness by handling different input types and managing errors efficiently.

CHAPTER 10: FUTURE SCOPE

The proposed Cognitive AI-Based Multi-Document Text Summarization with Visual Insight Extraction System provides an efficient and lightweight solution for document analysis. While the system successfully integrates text summarization, multi-format processing, and visual insight extraction, there are several opportunities for further enhancement and expansion to improve its capabilities and performance.

One of the major areas for future improvement is the integration of advanced deep learning models such as Transformer-based architectures (BERT, GPT, T5). These models can provide better contextual understanding and generate more accurate and human-like summaries. Incorporating such models can enhance the quality of summarization, especially for complex and domain-specific documents.

Another important enhancement is the implementation of abstractive summarization techniques. Currently, the system uses extractive methods that select sentences directly from the text. In the future, abstractive methods can be introduced to generate new sentences that better capture the meaning of the document, resulting in more natural and coherent summaries.

The system can also be extended to support multilingual processing. By integrating language translation models, the system can process and summarize documents in multiple languages. This would make the application more versatile and suitable for global users.

Further improvements can be made in the visual content analysis module. Currently, the system detects figures and tables using rule-based techniques. In the future, computer vision and deep learning methods can be used to analyze images, diagrams, and flowcharts more accurately. This would enable the system to generate more detailed and context-aware explanations of visual elements.

Another potential enhancement is the addition of speech and voice-based interaction. Users could upload audio files or use voice commands to interact with the system. This would improve accessibility and make the system more user-friendly.

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COGNITIVE AI-BASED MULTI-DOCUMENT TEXT SUMMARIZATION WITH VISUAL INSIGHT EXTRACTION USING LIGHTWEIGHT NLP TECHNIQUES

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***Abstract**: Text summarization has become an essential task in Natural Language Processing (NLP) to efficiently process and understand large volumes of textual data. This project presents a Cognitive AI-Based Text Summarization System that generates meaningful summaries from documents using intelligent text processing techniques. The system supports multiple input formats, including PDF files, images, and direct text input, making it versatile and user-friendly. It extracts textual content using PDF parsing and Optical Character Recognition (OCR) techniques and applies preprocessing steps such as cleaning and normalization to prepare the data for analysis. A frequency-based summarization approach is used to generate both page-wise summaries and a comprehensive final summary, ensuring important information is retained while reducing redundancy. In addition to text summarization, the system incorporates visual content analysis to identify and interpret figures and tables, providing meaningful insights to enhance document understanding. The implementation uses JavaScript, PDF.js, and Tesseract.js with a web-based interface, and experimental results demonstrate that the system effectively summarizes documents, handles multiple input formats, and produces accurate and concise outputs.*

***Keywords** — Text Summarization, NLP, Cognitive AI, PDF Processing, OCR, Visual Analysis, Document Summarization, Web Application*

1. INTRODUCTION

With the rapid growth of digital information, extracting meaningful insights from large textual documents has become a challenging task. Manual reading and analysis of documents such as research papers, reports, and articles is time-consuming and inefficient.

Text summarization, a key area of Natural Language Processing (NLP), aims to generate concise and meaningful summaries while preserving essential information. Traditional approaches often fail to capture context effectively or handle diverse input formats.

To address these challenges, this project proposes a **Cognitive AI-Based Text Summarization System** that supports multiple input formats such as PDF documents and images. The system integrates intelligent text processing techniques with visual content analysis to improve understanding and usability.

2. LITERATURE SURVEY

Existing research in text summarization highlights that extractive methods based on statistical techniques are simple and efficient but often fail to capture deep contextual meaning. These methods rely on frequency and sentence ranking, which may lead to loss of semantic understanding in complex documents.

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Advanced approaches using deep learning models such as LSTM and Transformer-based architectures improve summarization quality by learning contextual relationships within text. However, these models require large datasets, high computational resources, and are often difficult to interpret.

Recent systems also focus on document understanding by incorporating multimodal analysis, but most existing solutions lack the ability to handle multiple input formats such as PDFs and images effectively. Additionally, many systems do not provide insights into visual elements like figures and tables, limiting overall document comprehension.

This project addresses these limitations by implementing a hybrid approach that combines efficient frequency-based summarization with visual content analysis. It supports multi-format inputs using PDF parsing and OCR techniques while providing structured outputs such as page-wise summaries, final summaries, and visual insights. The system also emphasizes usability through an interactive web interface and improves overall document understanding.

3. PROPOSED SYSTEM

The proposed system is a **Cognitive AI-Based Text Summarization System** designed to overcome the limitations of traditional summarization approaches by integrating intelligent text processing with visual content analysis. The system combines frequency-based extractive summarization techniques with structured document analysis to generate accurate and meaningful summaries.

Unlike conventional systems that focus only on text summarization, the proposed system also analyzes visual elements such as figures and tables to enhance document understanding. It supports multiple input formats including PDF documents and images, making it more flexible and user-friendly.

The system leverages PDF parsing and Optical Character Recognition (OCR) to extract textual content from different sources. A frequency-based scoring mechanism is used to identify important sentences, ensuring efficient and interpretable summarization. Additionally, visual detection techniques are applied to extract captions and provide meaningful explanations for figures and tables.

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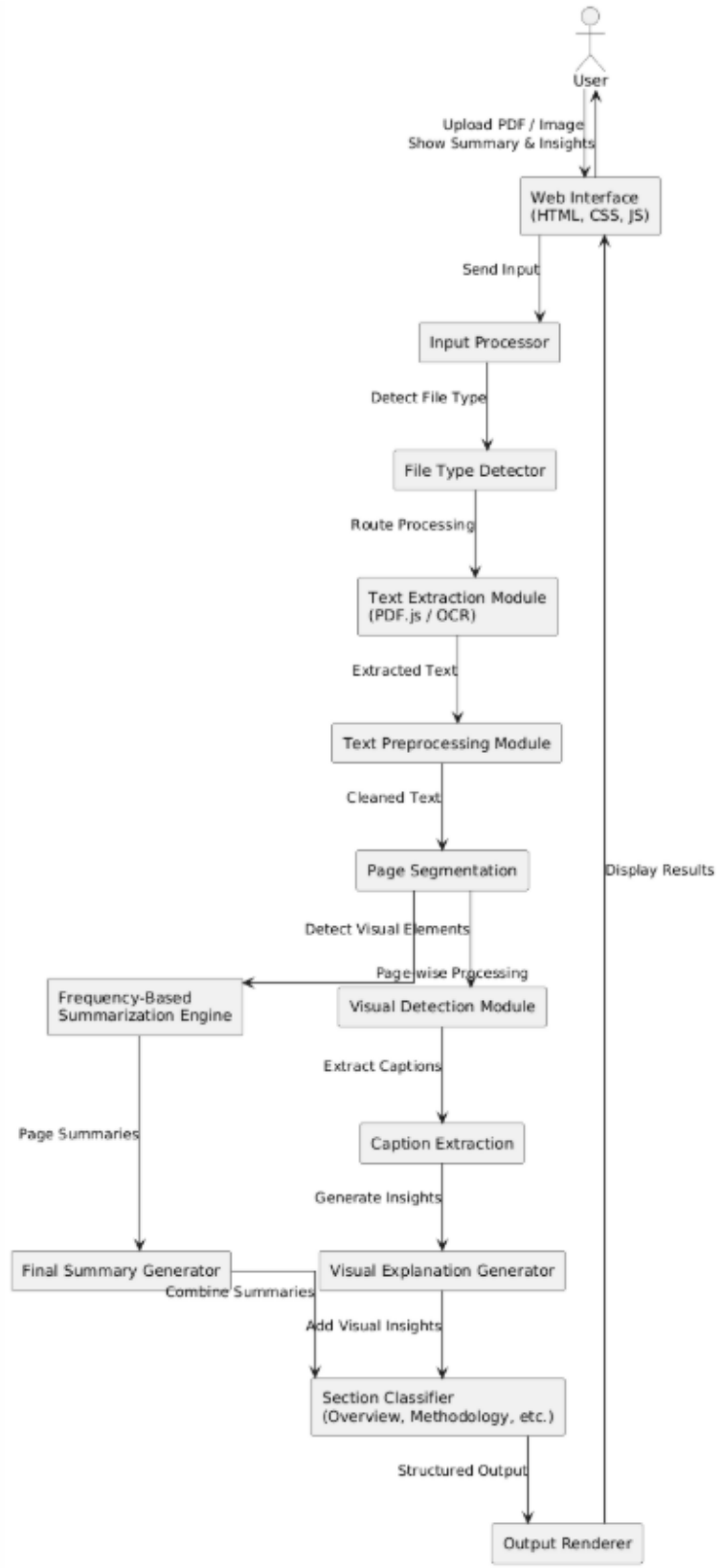

Fig. 1: Proposed System

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As shown in Fig. 1, the system follows a structured workflow that integrates text extraction, preprocessing, summarization, and visual content analysis to generate meaningful outputs.

Key Features

• **Multi-format input support:** The system accepts PDF files and images, enabling users to process various types of documents.
• **Page-wise summarization:** Generates summaries for each page individually to provide detailed insights.
• **Final cognitive summary:** Produces an overall summary by combining important information from all pages.
• **Visual content analysis:** Detects figures and tables and provides descriptive insights.
• **Text extraction using PDF.js and OCR:** Ensures accurate extraction from both digital and scanned documents.
• **Noise removal and preprocessing:** Cleans unwanted content such as repeated headers or irrelevant text.
• **Interactive web interface:** Provides a user-friendly platform for uploading files and viewing results.
• **Efficient summarization algorithm:** Uses frequency-based scoring for fast and reliable summary generation.

Working of the System

The system follows a structured pipeline:

1. The user uploads a document (PDF or image) through the web interface.
2. The system processes the input and identifies the file type.
3. Text is extracted using PDF.js for PDFs or OCR techniques for images.
4. The extracted text is cleaned by removing noise and unnecessary content.
5. The document is divided into pages and processed individually.
6. A frequency-based summarization algorithm is applied to generate page-wise summaries.
7. Visual elements such as figures and tables are detected using pattern matching techniques.
8. Captions are extracted and meaningful explanations are generated for visual content.
9. A final cognitive summary is created by combining key information from all pages.
10. The system organizes summaries into structured sections such as Overview, Methodology, Results, and Conclusion.
11. The results, including page-wise summaries, final summary, and visual insights, are displayed to the user through the interface.

4. METHODOLOGY

The methodology of the proposed system is based on an intelligent text processing approach that combines extractive summarization techniques with visual content analysis. The system processes documents through multiple stages including text extraction, preprocessing, summarization, and visual interpretation to generate meaningful outputs.

4.1 Text Extraction

The text extraction module is responsible for retrieving textual content from different input formats such as PDF files and images.

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During processing:
• PDF documents are processed using PDF.js to extract text content
• Image-based documents are processed using Optical Character Recognition (OCR)
• Extracted text is arranged in proper reading order
During execution:
• The system identifies the file type
• Appropriate extraction technique (PDF or OCR) is applied
• Text from each page is stored for further processing
Advantages:
• Supports multiple input formats
• Accurate extraction from digital and scanned documents
• Enables flexible document handling
Limitation:
• OCR accuracy depends on image quality
• Complex layouts may affect extraction order

4.2 Text Preprocessing and Summarization
This module focuses on cleaning and summarizing the extracted text using a frequency-based approach.
Working:
• Text is cleaned by removing noise such as repeated headers and irrelevant content
• Sentences are extracted and filtered based on length and relevance
• Word frequency is calculated to identify important terms
• Sentences are scored based on frequency
• Top-ranked sentences are selected as summaries
The summarization process includes:
• Page-wise summary generation
• Final summary generation by combining key content
Advantages:
• Fast and computationally efficient
• Easy to implement and interpret
• Reduces redundancy in text
Limitation:
• Does not fully capture deep semantic meaning
• Depends on frequency rather than context

4.3 Visual Content Analysis
The system includes a visual analysis module to enhance document understanding by detecting and interpreting figures and tables.
Working:
• Pattern matching techniques are used to detect figures and tables
• Captions are extracted from the document
• Descriptive explanations are generated for each visual element
• Duplicate visual detections are filtered
Features:
• Identifies figures and tables from text patterns
• Provides meaningful explanations for visuals
• Enhances overall document comprehension
Advantages:
• Improves understanding of graphical content
• Adds contextual insights beyond text
• Supports academic and technical documents

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Limitation:
• Depends on proper caption formatting
• Limited handling of complex visual layouts

4.4 Section Classification and Output Generation

This module organizes the summarized content into structured sections and presents the final output.
Working:
• The combined text is divided into sections such as Overview, Methodology, Results, and Conclusion
• Summaries are generated for each section
• Page-wise summaries and visual insights are integrated
• Results are displayed through an interactive interface
Advantages:
• Provides structured and readable output
• Improves user experience
• Enables quick understanding of documents

This methodology ensures that the system efficiently processes documents, generates accurate summaries, and enhances understanding through visual content analysis. The integration of multiple modules makes the system robust, scalable, and suitable for real-world applications.

5. RESULTS AND DISCUSSION

The proposed system was evaluated using multiple documents, including research papers, reports, and mixed-format files containing both text and visual elements. The results demonstrate the effectiveness of the system in generating accurate summaries and extracting meaningful insights.

Summarization Scenarios

• **Text-rich pages:**
Pages containing sufficient textual content produce clear and informative summaries. Important sentences are accurately identified using frequency-based scoring, resulting in high-quality page-wise summaries.

• **Moderate content pages:**
Pages with moderate text generate summaries that capture key points but may include limited contextual detail. The summaries remain relevant but slightly less comprehensive.

• **Visual-dominant pages:**
Pages with figures, tables, or diagrams and minimal text are identified as visual pages. The system generates descriptive insights based on detected captions, ensuring meaningful interpretation even when textual data is limited.

• **Noisy or irregular content:**
Pages containing unwanted elements such as repeated headers or formatting noise are cleaned during preprocessing. The system effectively removes irrelevant content and produces readable summaries.

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System Performance

The system successfully:

• Generated accurate page-wise and final summaries for multi-page documents
• Extracted text effectively from PDF files using PDF.js
• Handled visual content by detecting figures and tables and providing explanations
• Removed noise such as repeated or irrelevant text during preprocessing
• Processed multiple pages efficiently with consistent performance
• Displayed structured outputs including summaries and visual insights through an interactive interface

Discussion

The results indicate that the proposed system performs effectively across different document types and content scenarios.

• The frequency-based summarization method works well for extracting important information from text-heavy documents.
• The visual analysis module enhances document understanding by interpreting graphical elements.
• The combination of text summarization and visual insight generation improves overall output quality.

The system successfully handles various cases including text-rich pages, moderate content, visual-dominant pages, and noisy data. The structured output format further improves readability and usability, making the system suitable for academic and real-world applications.

Overall, the system demonstrates reliable performance, efficiency, and scalability in summarizing documents and providing meaningful insights.

6. CONCLUSION

The proposed **Cognitive AI-Based Text Summarization System** successfully addresses the limitations of traditional summarization techniques by integrating efficient text processing with visual content analysis. The system generates accurate, concise, and structured summaries, making it suitable for real-world document analysis applications.
The inclusion of multi-format input support enhances usability, allowing users to process PDF documents and images seamlessly. The use of PDF parsing and OCR techniques ensures effective text extraction from both digital and scanned documents.

A significant contribution of this work is the incorporation of visual content analysis. By detecting figures and tables and providing meaningful explanations, the system enhances overall document understanding beyond plain text summarization.
The frequency-based summarization approach ensures fast and reliable performance, while preprocessing techniques improve the quality of extracted content by removing noise and irrelevant data. The system also presents outputs in a structured format, including page-wise summaries, final summaries, and visual insights, improving readability and user experience.

Overall, the system demonstrates strong performance, efficiency, and scalability. It can be further extended by integrating advanced deep learning models, supporting multilingual summarization, and enabling real-time applications such as automated report analysis, research paper summarization, and intelligent document processing systems.

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